

Q1:WRITE A PROGRAM THAT ACCEPTS THE MARKS OF FIVE SUBJECTS AND FIND SUM AND PERCENTAGE MARKS OBTAINDE BY STUDENT.

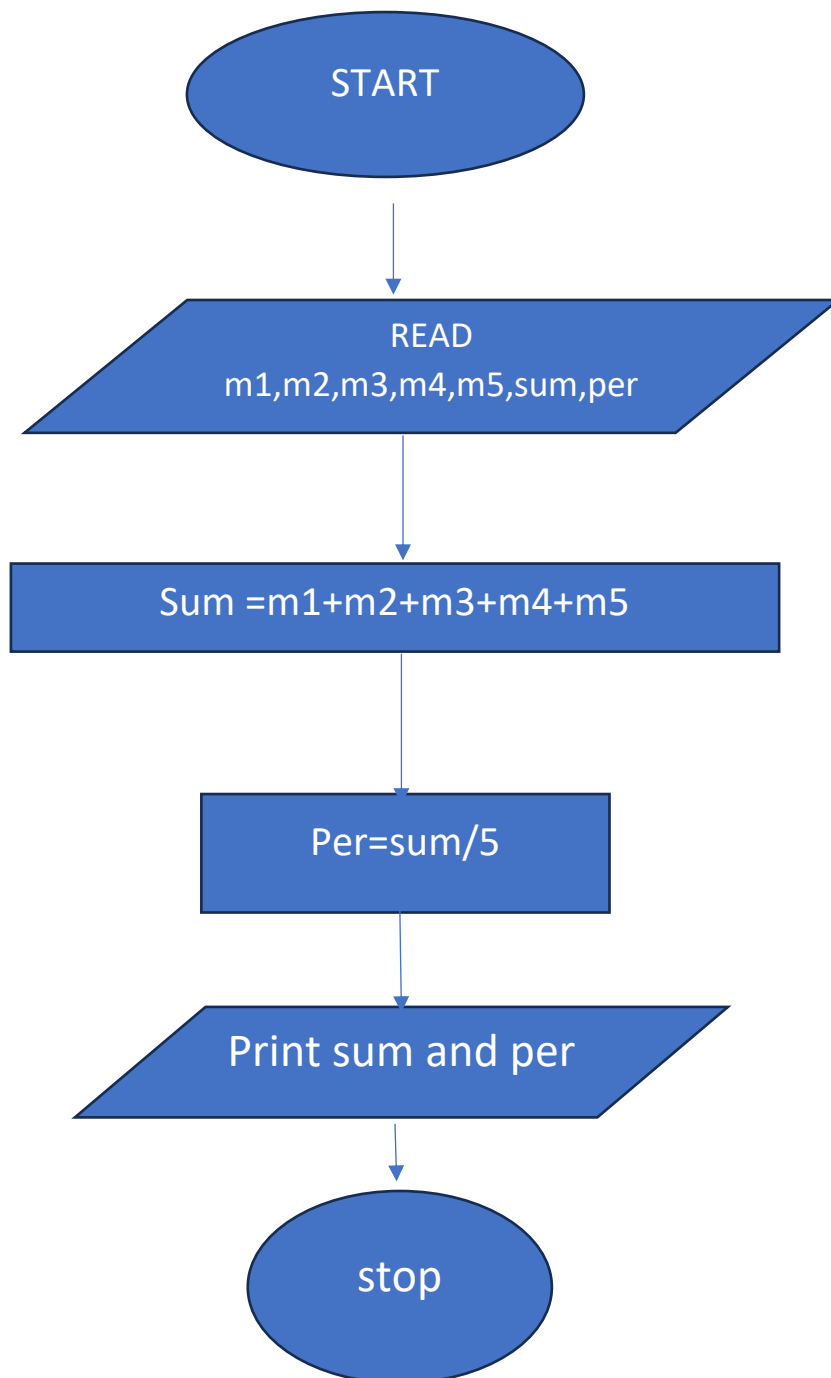
```
#include<stdio.h>
#include<conio.h>
Void main()
{
float m1,m2,m3,m4,m5,sum,per;
clrscr();
printf("Enter the marks of five subjects:");
scanf("%f%f%f%f%f",&m1,&m2,&m3,&m4,&m5);
sum=m1+m2+m3+m4+m5;
per=sum/5;
printf("Sum of five subjects are %f\n",sum);
printf("Percentage of five subjects are %f",per);
getch();
}
```

OUTPUT

```
Enter the marks of five subjects:  
90  
95  
90  
95  
80  
Sum of five subjects are 450.000000  
Percentage of five subjects are 90.000000
```

ALGORITHM AND FLOWCHART

- Read five subject marks and store them into 5 different variables.
- Calculate the sum of all subjects and store in total
 $\text{total} = m1 + m2 + m3 + m4 + m5$.
- Divide the sum of all subjects by the total number of subjects to find the average i.e. $\text{average} = \text{total} / 5$.
- Calculate the percentage using $\text{percentage} = (\text{total} / 500) * 100$.
- Finally, print resultant values total and percentage.



Q2:WRITE A PROGRAM THAT CALCULATES SIMPLE INTEREST AND COMPOUND INTEREST .THE PRINCIPAL AMOUNT ,RATE OF INTEREST AND TIME ARE ENTERED THROUGH KEYBOARD.

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
Void main()
{
float p,r,t,n,si,ci;
clrscr();

printf("Enter the principal amount ,rate of
interest,time,number of times interest is
compounded per year:");

scanf("%f%f%f%f",&p,&r,&t,&n);

si=(p*r*t)/100;

ci=p*pow((1+r/n),n*t)-p;

printf("The value of simple interest is %f\n",si);
printf(The value of compound interest is %f\n",ci);
```

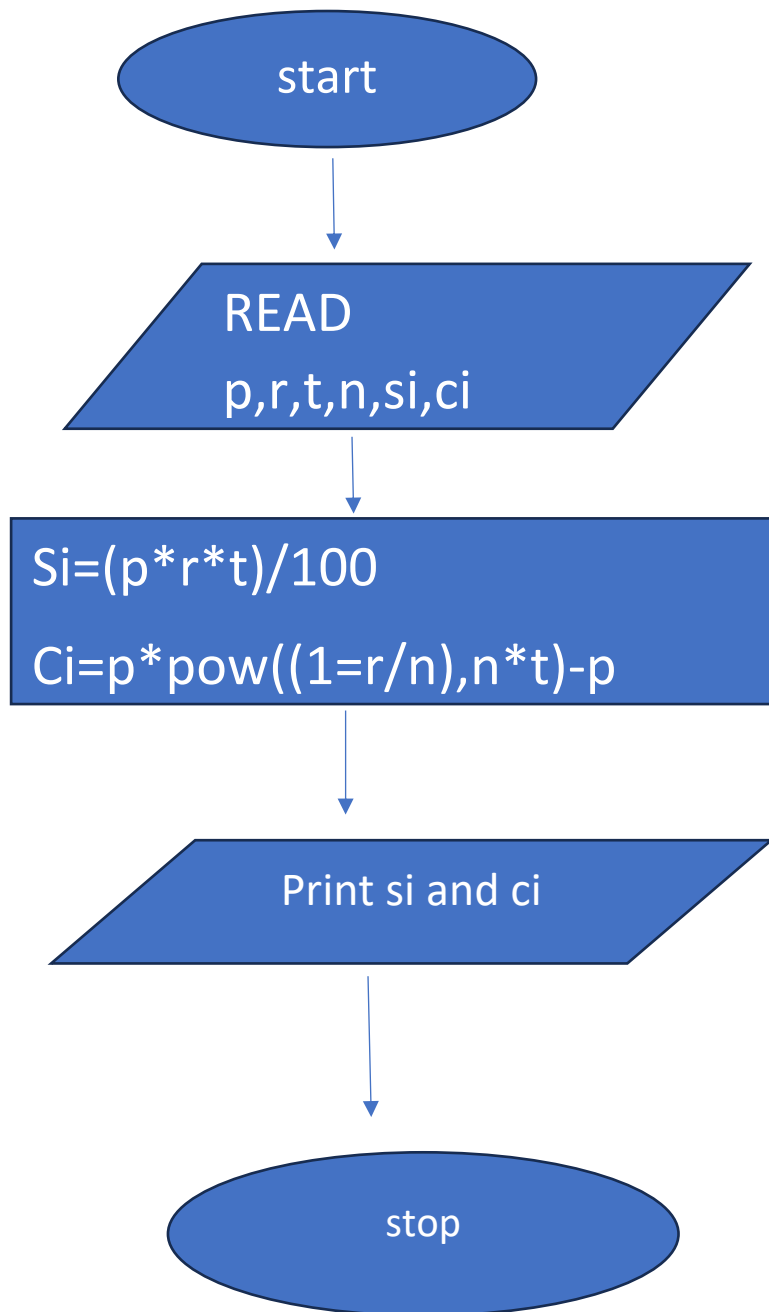
```
getch();  
}
```

OUTPUT

```
Enter the pricipal amount,rate of interest,time,number of times the interest is  
compunded per year:1000  
5  
6  
1  
The value of simple interest is 300.000000  
The value of compound interest is 46655000.000000
```

ALGORITHM AND FLOWCHART

- 1.START
- 2.READ p,r,t,n,si,ci
3. $Si = (p * r * t) / 100$
4. $Ci = p * \text{pow}((1 + r/n), n * t) - p$
- 5.Print si and ci
- 6.Stop



Q3:WRITE A PROGRAM TO CALCULATE AREA AND CIRCUMFERENCE OF A CIRCLE.

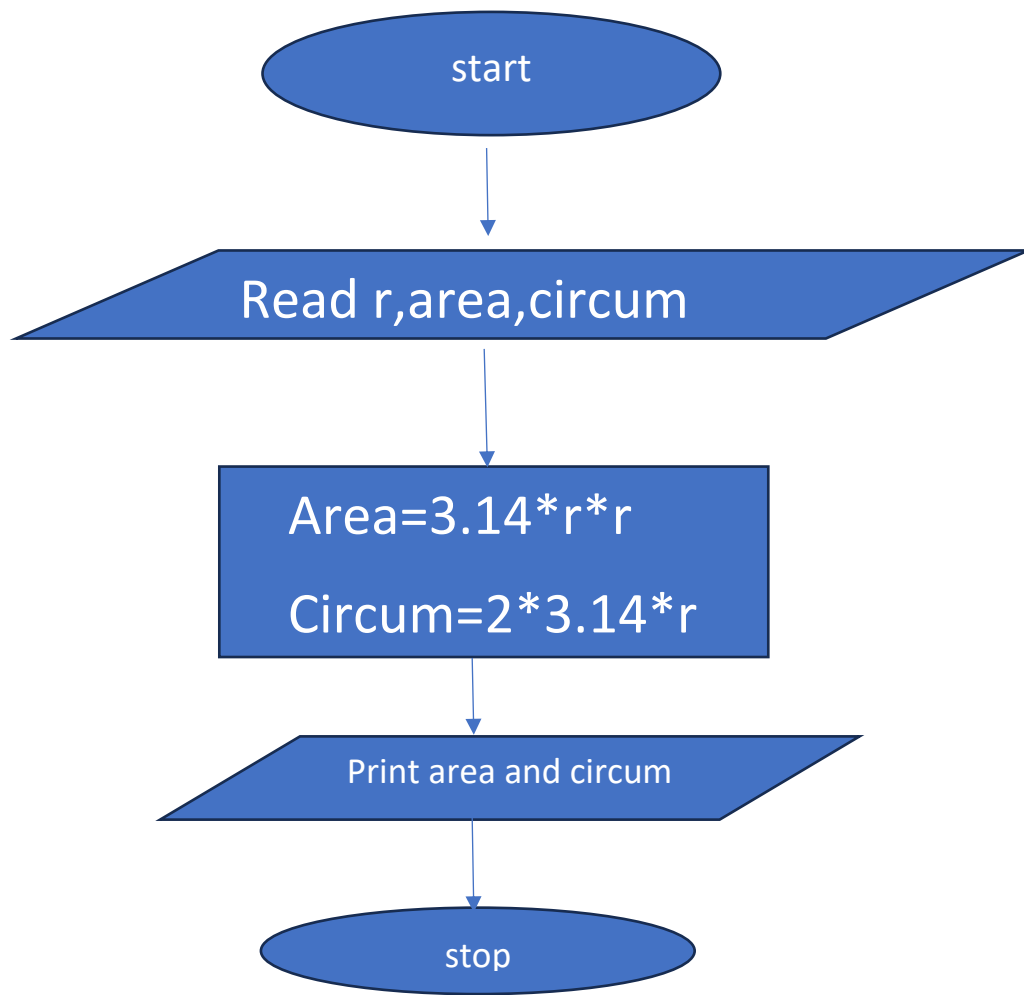
```
#include<stdio.h>
#include<conio.h>
Void main()
{
float r,area,circum;
printf("Enter the radius:");
scanf("%f",&r);
area=3.14*r*r;
circum=2*3.14*r;
printf("The area and circumference is %f and
%f",area,circum");
getch();
}
```

OUTPUT

```
Enter the radius:5  
The area and circumference of circle is 78.500000 and 31.400000_
```

ALGORITHM AND FLOWCHART

- 1.Start
- 2.Read r,area,circum
- 3.Area= $3.14*r*r$
- 4.Circum= $2*3.14*r$
- 5.Printarea and circum
- 6.Stop



Q4:WRITE A PROGRAM THAT ACCEPTS TEMPERATURE IN CENTIGRADE AND CONVERT INTO FAHRENHEIT.

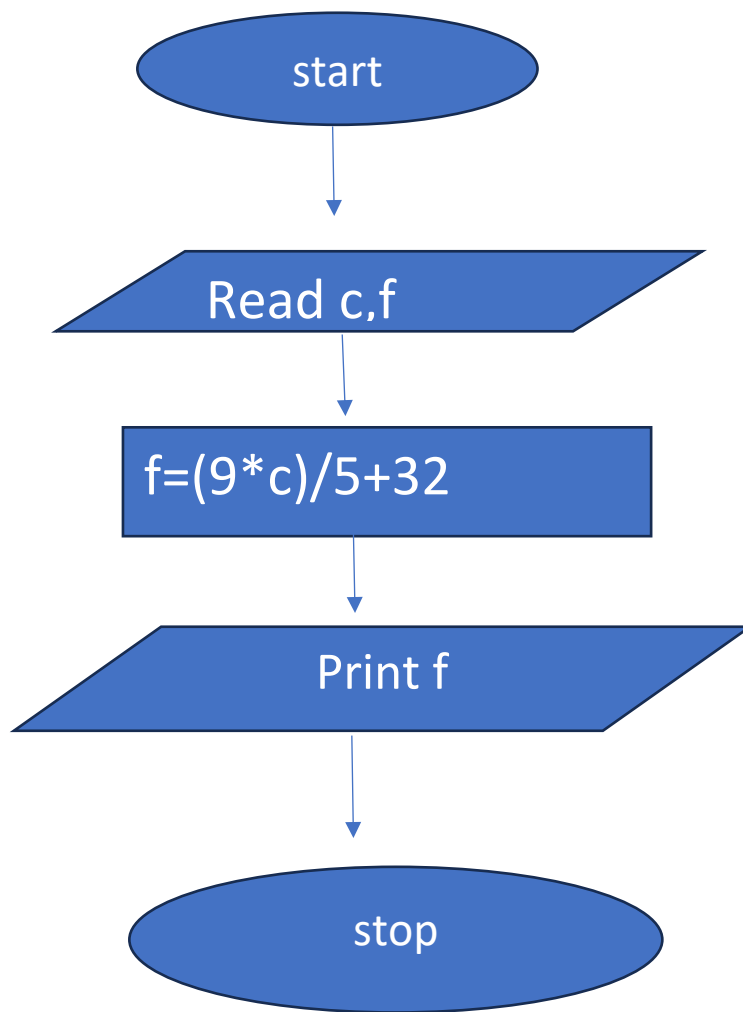
```
#include<stdio.h>
#include<conio.h>
void main()
{
float c,f;
clrscr();
printf("Enter the value of temperature in
centigrade:\n");
scanf("%f",&c);
f=(9*c)/5+32;
printf("Value of temperature in fahrenheit is
%f:",f);
getch();
}
```

OUTPUT

```
Enter the value of temperature in centigrade:  
16  
Value of temperature in fahrenheit is 60.799999:
```

ALGORITHM AND FLOWCHART

- 1.Start
- 2.Read c,f
3. $f = (9 * c) / 5 + 32$
- 4.Printf f
- 5.Stop



Q5: a: WRITE A PROGRAM THAT SWAPS VALUES OF TWO VARIABLES USING A THIRD VARIABLE.

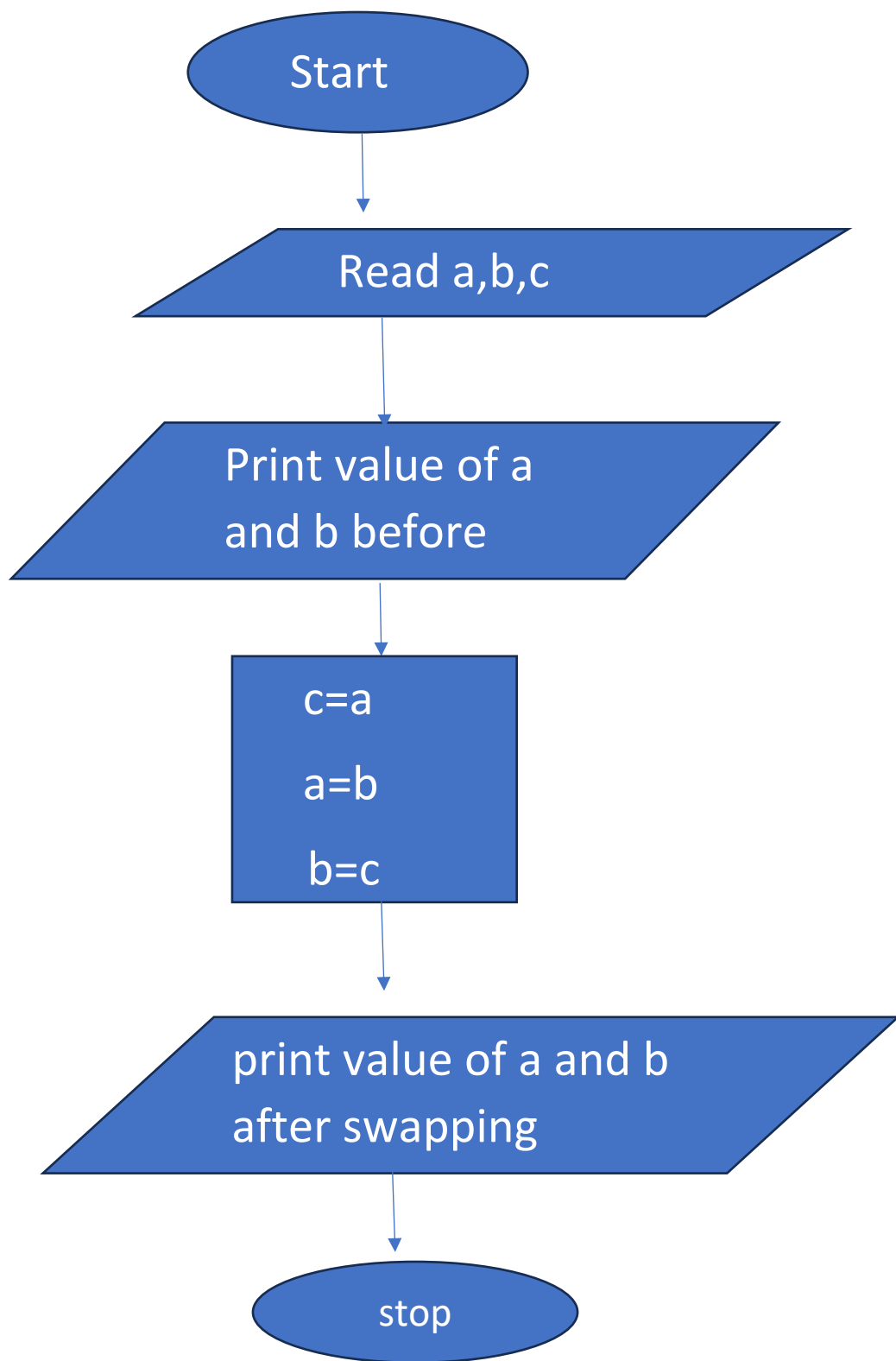
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c;
clrscr();
printf("Enter values of a and b:");
scanf("%d%d",&a,&b);
printf("Value of a and b before swapping are %d
and %d\n",a,b);
c=a;
a=b;
b=c;
printf("Value of a and b after swapping are %d
and %d",a,b);
getch();
}
```

OUTPUT

```
Enter values of a and b:5
6
Value of a and b before swapping are 5 and 6
Value of a and b after swapping are 6 and 5_
```

ALGORITHM AND FLOWCHART

- 1.Start
- 2.Read a,b,c
- 3.Print value of a and b before swapping
- 4.c=a
a=b
b=c
- 5.print value of a and b after swapping
- 6.stop



Q5:b: WRITE A PROGRAM THAT SWAPS VALUES OF TWO VARIABLES.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c;
clrscr();
printf("Enter values of a and b:");
scanf("%d%d",&a,&b);
printf("Value of a and b before swapping are %d
and %d\n",a,b);
a=a+b;
b=a-b;
a=a-b;
printf("Value of a and b after swapping are %d
and %d",a,b);
getch();
}
```

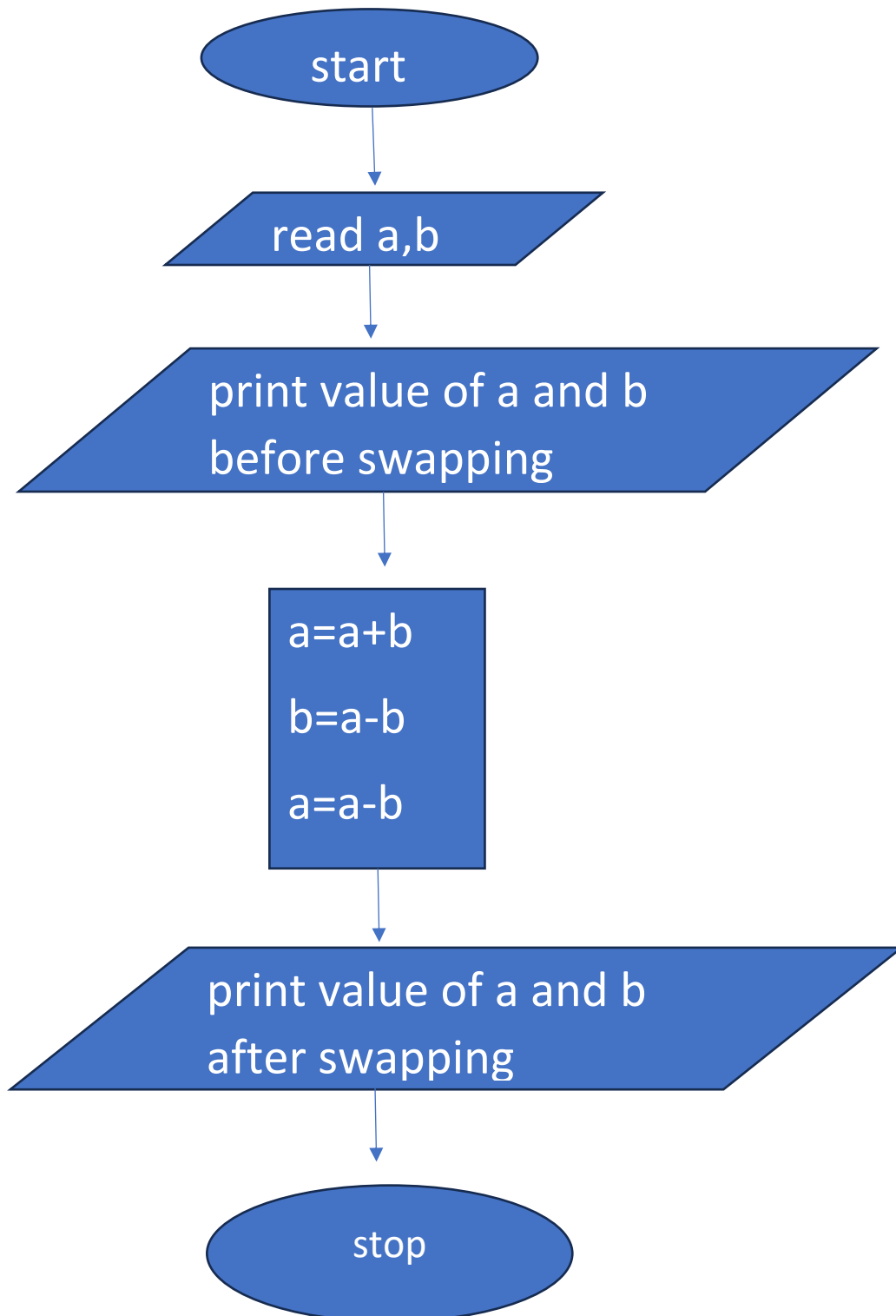
OUTPUT

```
Enter values of a and b:9
4
Value of a and b before swapping are 9 and 4
Value of a and b after swapping are 4 and 9
```

ALGORITHM

- 1.start
- 2.read a,b
- 3.print value of a and b before swapping
4. $a=a+b$
 $b=a-b$
 $a=a-b$
- 5.print value of a and b after swapping
- 6.stop

FLOWCHART



Q6:WRITE A PROGRAM THAT CHECKS WHETHER THE TWO NUMBERS ENTERED BY THE USER ARE EQUAL OR NOT.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int num1,num2;
clrscr();
printf("Enter the number1 and number2:");
scanf("%d%d",&num1,&num2);
if(num1==num2)
{
printf("%d is equal to %d\n",num1,num2);
}
else
{
printf("%d is not equal to %d",num1,num2);
```

```
}  
getch();  
}
```

OUTPUT

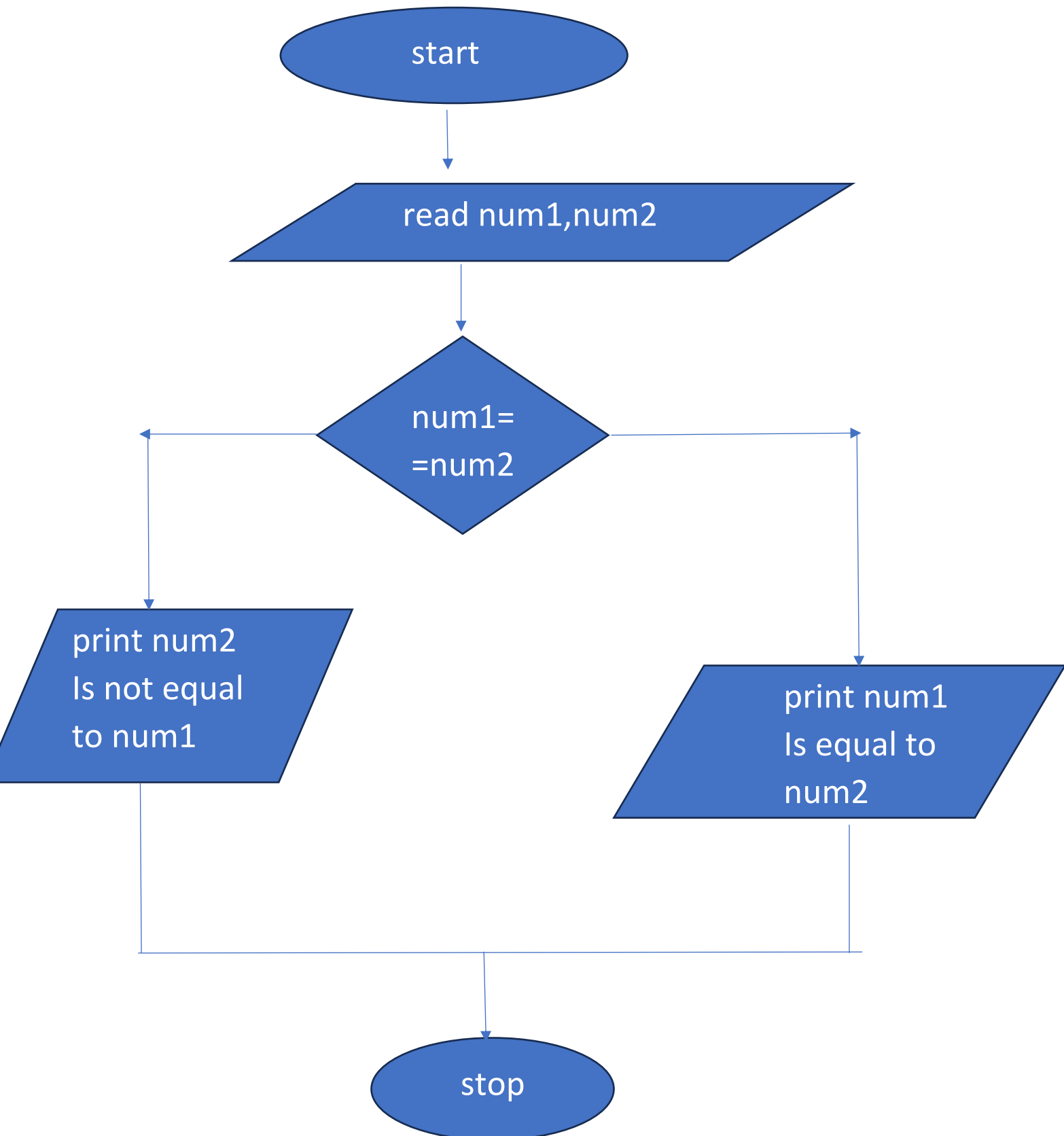
```
Enter the number1 and number2:7  
4  
7 is not equal to 4_
```

```
Enter the number1 and number2:8  
8  
8 is equal to 8
```

ALGORITHM

1. start
2. read num1,num2
3. if num1==num2
 print num1 is equal to num2
 else
 print num2 is not equal to num1
4. stop

FLOWCHART

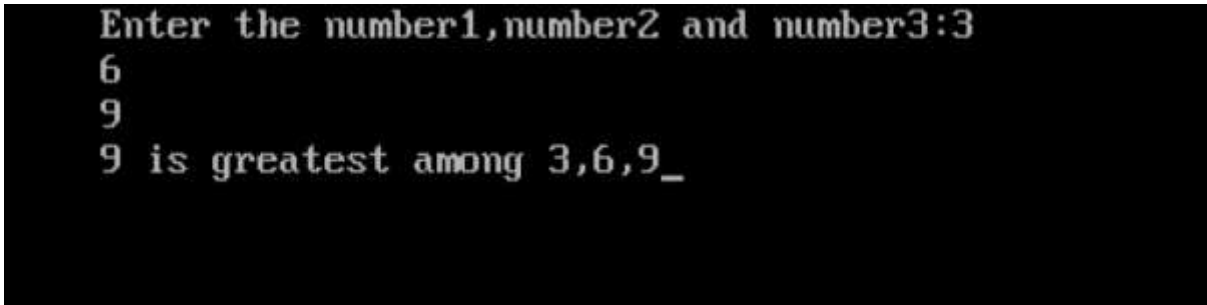


Q7:WRITE APROGRAM TO FIND GREATEST OF THREE NUMBERS.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int num1,num2,num3;
clrscr();
printf("Enter the number1,number2 and
number3:");
scanf("%d%d%d",&num1,&num2,&num3);
if((num1>num2)&&(num1>num3))
{
printf("%d is greatest among
%d,%d,%d",num1,num1,num2,num3);
}
else if((num2>num1)&&(num2>num3))
```

```
{  
    printf("%d is greatest among  
%d,%d,%d",num2,num1,num2,num3);  
}  
else  
{  
    printf("%d is greatest among  
%d,%d,%d",num3,num1,num2,num3);  
}  
getch();  
}
```

OUTPUT

A screenshot of a terminal window with a black background and white text. It shows the execution of a C program that finds the greatest of three numbers. The prompt 'Enter the number1,number2 and number3:3' is followed by the user input '6' on the next line and '9' on the third line. The final output line is '9 is greatest among 3,6,9_'.

```
Enter the number1,number2 and number3:3  
6  
9  
9 is greatest among 3,6,9_
```

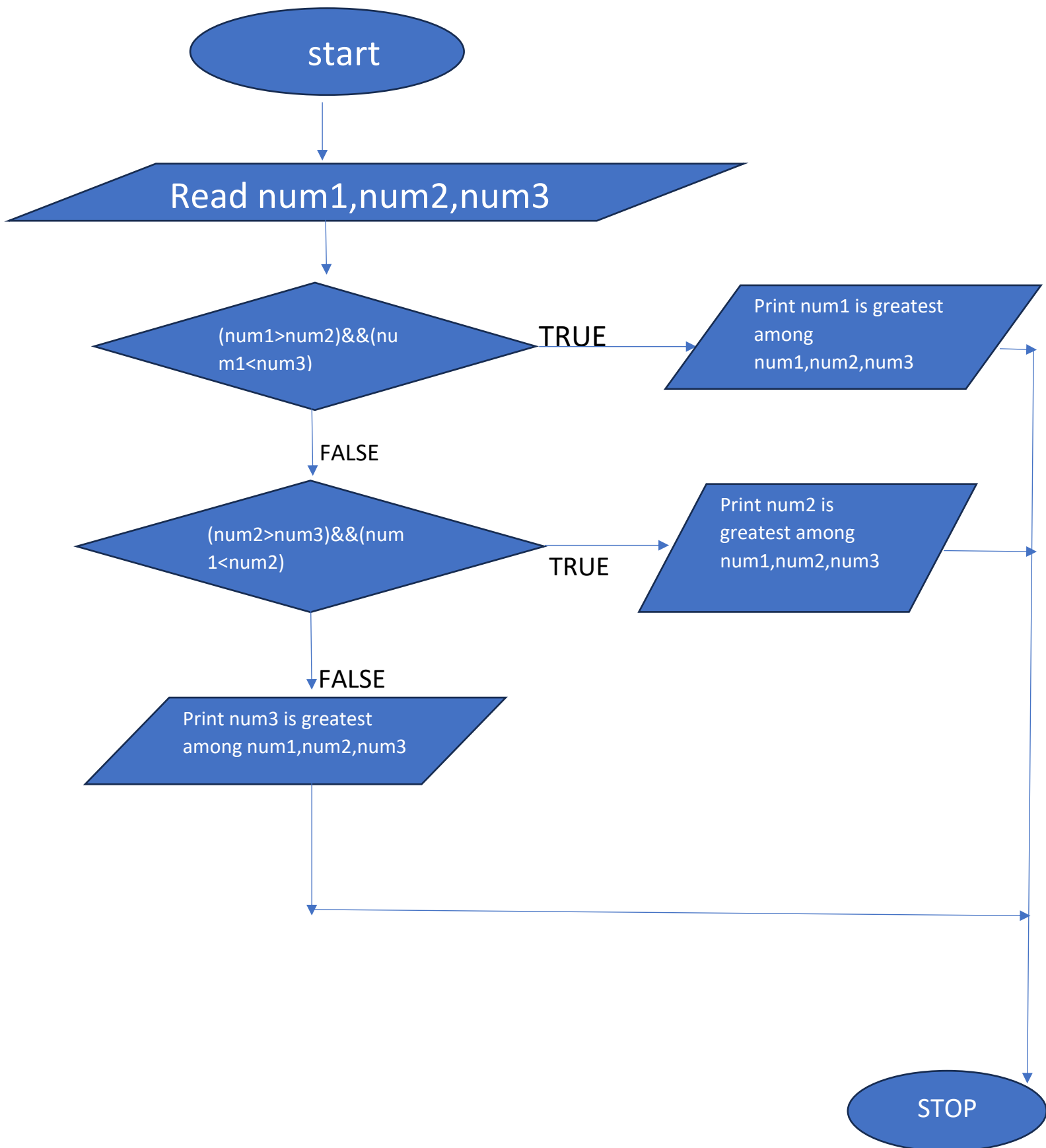
```
Enter the number1,number2 and number3:6
9
3
9 is greatest among 6,9,3
```

```
Enter the number1,number2 and number3:9
3
6
9 is greatest among 9,3,6
```

ALGORITHM

- start
- read num1,num2,num3
- if (num1>num2)&&(num1<num3)
print num1 is greatest among
num1,num2,num3
- else if (num2>num3)&&(num2>num1)
print num2 is greatest among
num1,num2,num3
- else
print num3 is greatest among
num1,num2,num3
- stop

FLOWCHART

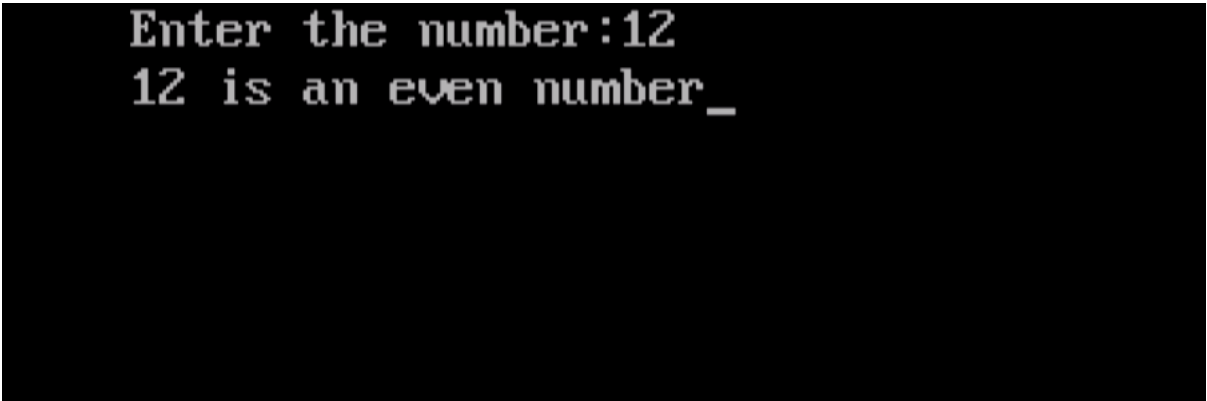


Q8:WRITE A PROGRAM THAT FINDS WHETHER A NUMBER IS EVEN OR ODD.

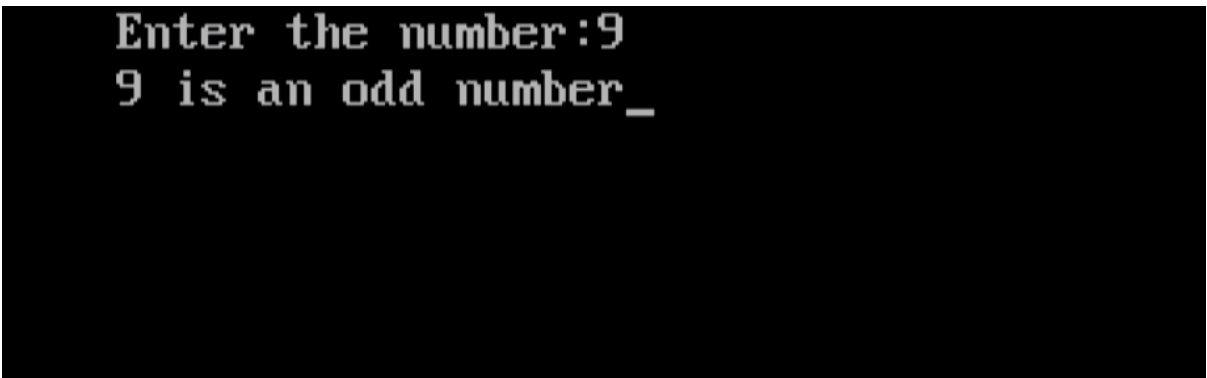
```
#include<stdio.h>
#include<conio.h>
void main()
{
int num;
clrscr();
printf("Enter the number:");
scanf("%d",&num);
if(num%2==0)
{
printf("%d is an even number",num);
}
else
{
printf("%d is an odd number",num);
}
```

```
getch();  
}
```

OUTPUT



```
Enter the number:12  
12 is an even number_
```



```
Enter the number:9  
9 is an odd number_
```

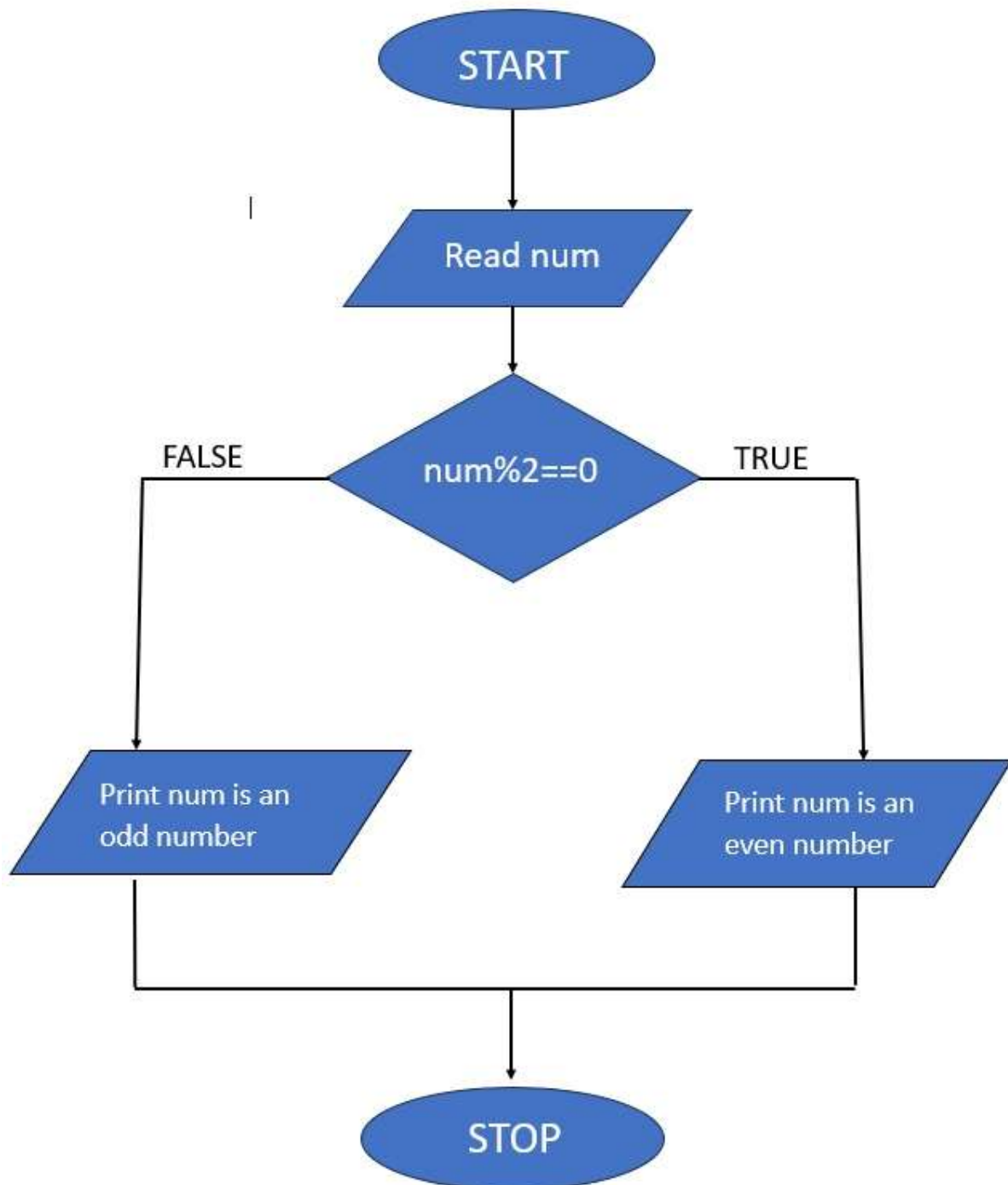
ALGORITHM

- start
- read num
- if $\text{num} \% 2 == 0$
 - print num is an even number
 - else

print num is an odd number

➤ stop

FLOWCHART

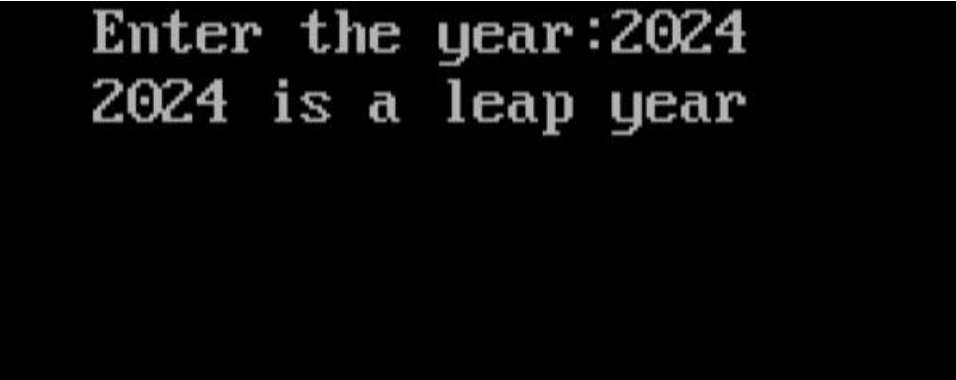


Q9:WRITE A PROGRAM THAT TELLS WHETHER A GIVEN YEAR IS A LEAP YEAR OR NOT.

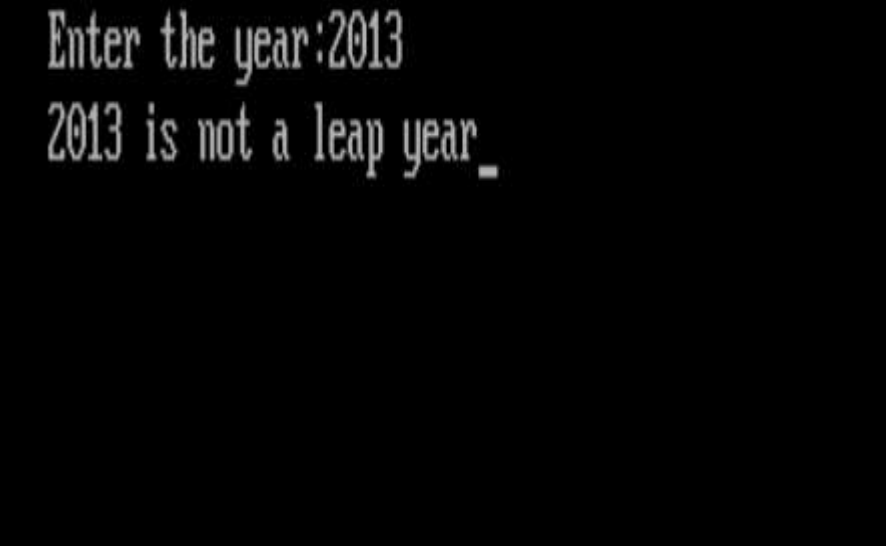
```
#include<stdio.h>
#include<conio.h>
void main()
{
int year;
clrscr();
printf("Enter the year:");
scanf("%d",&year);
if((year%400==0) || (year%4==0)&&(year%100!=0)
)
{
printf("%d is a leap year",year);
}
else
{
printf("%d is not a leap year",year);
```

```
}  
getch();  
}
```

OUTPUT



```
Enter the year:2024  
2024 is a leap year
```

A terminal window with a black background and white text. The first line shows the prompt "Enter the year:" followed by the input "2024". The second line shows the output "2024 is a leap year".

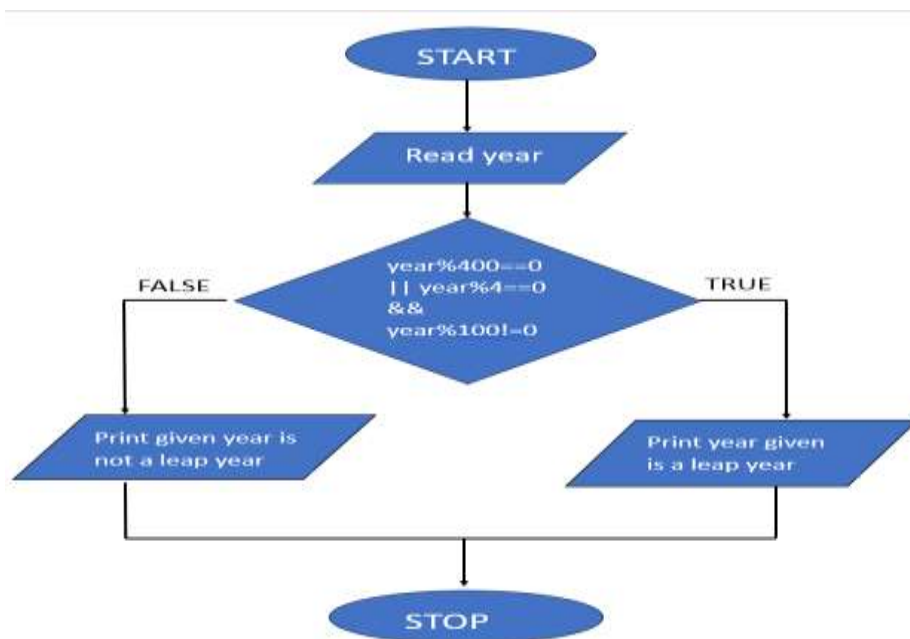
```
Enter the year:2013  
2013 is not a leap year_
```

A terminal window with a black background and white text. The first line shows the prompt "Enter the year:" followed by the input "2013". The second line shows the output "2013 is not a leap year_".

ALGORITHM

- start
- read year
- if $(\text{year} \% 400 == 0) \parallel (\text{year} \% 4 == 0 \ \&\& \text{year} \% 100 != 0)$
 print year given is a leap year
 else
 print given year is not a leap year
- stop

FLOWCHART

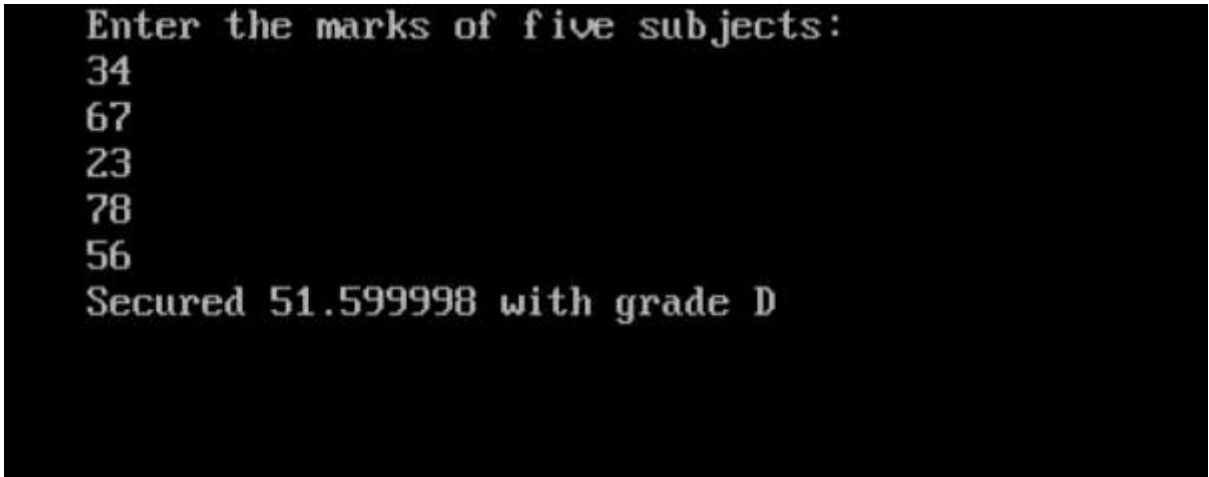


Q10:WRITE A PROGRAM THAT ACCEPTS MARKS OF FIVE SUBJECTS AND FINDS PERECNTAGE AND PRINTS GRADE .

```
#include<stdio.h>
#include<conio.h>
void main()
{
float m1,m2,m3,m4,m5,sum,per;
clrscr();
printf("Enter the marks of five subjects:\n");
scanf("%f%f%f%f%f",&m1,&m2,&m3,&m4,&m5);
sum=m1+m2+m3+m4+m5;
per=sum/5;
if(per>=90)
{
printf("Secured %f with grade A",per);
}
else if(per>=80&& per<90)
```

```
{  
    printf("Secured %f with grade B",per);  
}  
else if(per>=60 && per<80)  
{  
    printf("Secured %f with grade C",per);  
}  
else  
{  
    printf("Secured %f with grade D",per);  
}  
getch();  
}
```

OUTPUT

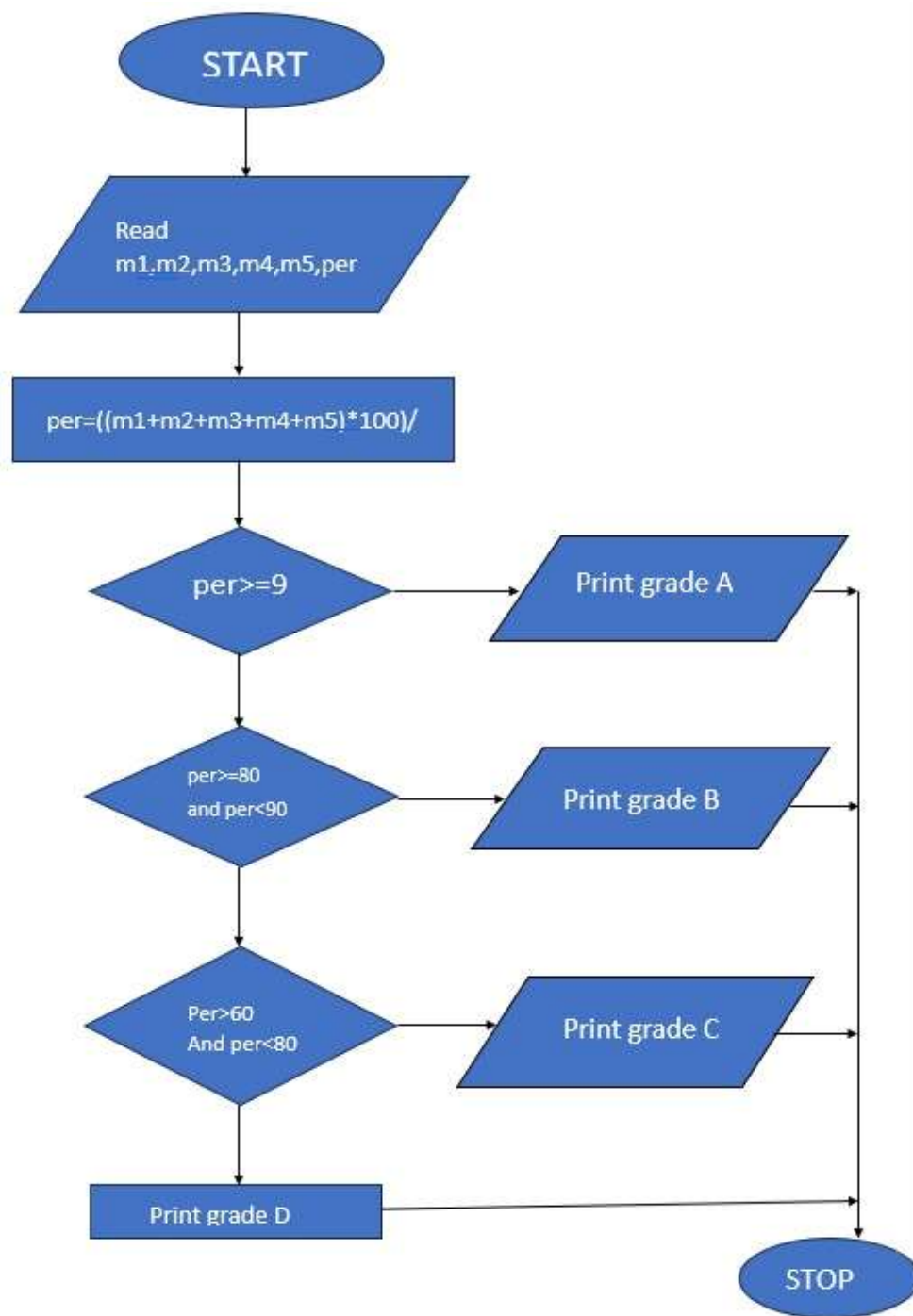
A screenshot of a terminal window with a black background and light green text. It shows the execution of a C program. The prompt 'Enter the marks of five subjects:' is followed by five lines of input: 34, 67, 23, 78, and 56. The final output line is 'Secured 51.599998 with grade D'.

```
Enter the marks of five subjects:  
34  
67  
23  
78  
56  
Secured 51.599998 with grade D
```

ALGORITHM

- start
- read m1,m2,m3,m4,m5,per
- $per = ((m1 + m2 + m3 + m4 + m5) * 100) / 500$;
- if $per \geq 90$
print secured % with grade A
- else if $per \geq 80$ and $per < 90$
print grade B
- else if $per \geq 60$ and $per < 80$
print grade C
- else
print grade D
- stop

FLOWCHART



Q11:WRITE A PROGRAM THAT TAKES TWO OPERANDS AND ONE OPERATOR FROM THE USER AND PERFORM THE OPERATION AND PRINTS RESULT BY USING SWITCH STATEMENT.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int num1,num2,op;
printf("Enter the two numbers:");
scanf("%d%d",&num1,&num2);
printf("Enter the operator:\n 1 for addition \n 2
for subtraction\n 3for multiplication \n 4 for
division\n 5 for modulo:\n");
scanf("%d",&op);
switch(op)
{
case 1:
printf("%d+%d=%d\n",num1,num2,num1+num2);
```

```
break;
```

```
case 2:
```

```
printf("%d-%d=%d\n",num1,num2,num1-num2);
```

```
break;
```

```
case 3:
```

```
printf("%d*%d=%d\n",num1,num2,num1*num2);
```

```
break;
```

```
case 5:
```

```
printf("%d%%%d=%d\n",num1,num2,num1%num  
2);
```

```
break;
```

```
default:
```

```
printf("You made a wrong choice");
```

```
break;
```

```
}
```

```
getch();
```

```
}
```

OUTPUT

```
C:\TURBOC3\BIN>TC
Enter the two numbers:8
4
Enter the operator:
1 for addition
2 for subtraction
3for multiplication
4 for division
5 for modulo:
1
8+4=12
```

ALGORITHM

1. Start
2. Read num1, num2, op
3. Print enter 1 for addition \n 2 for subtraction \n 3 for multiplication \n 4 for division \n 4 for division \n 5 for modulo
4. Switch(op)
 - Case 1
 - Print addition of num 1 and num 2
 - and break

Case 2

Print subtraction of num1 and num2
and break

Case 3

Print multiplication of num1 and
break

Case 4

Print division of num 1 and num 2
and break

Case 5

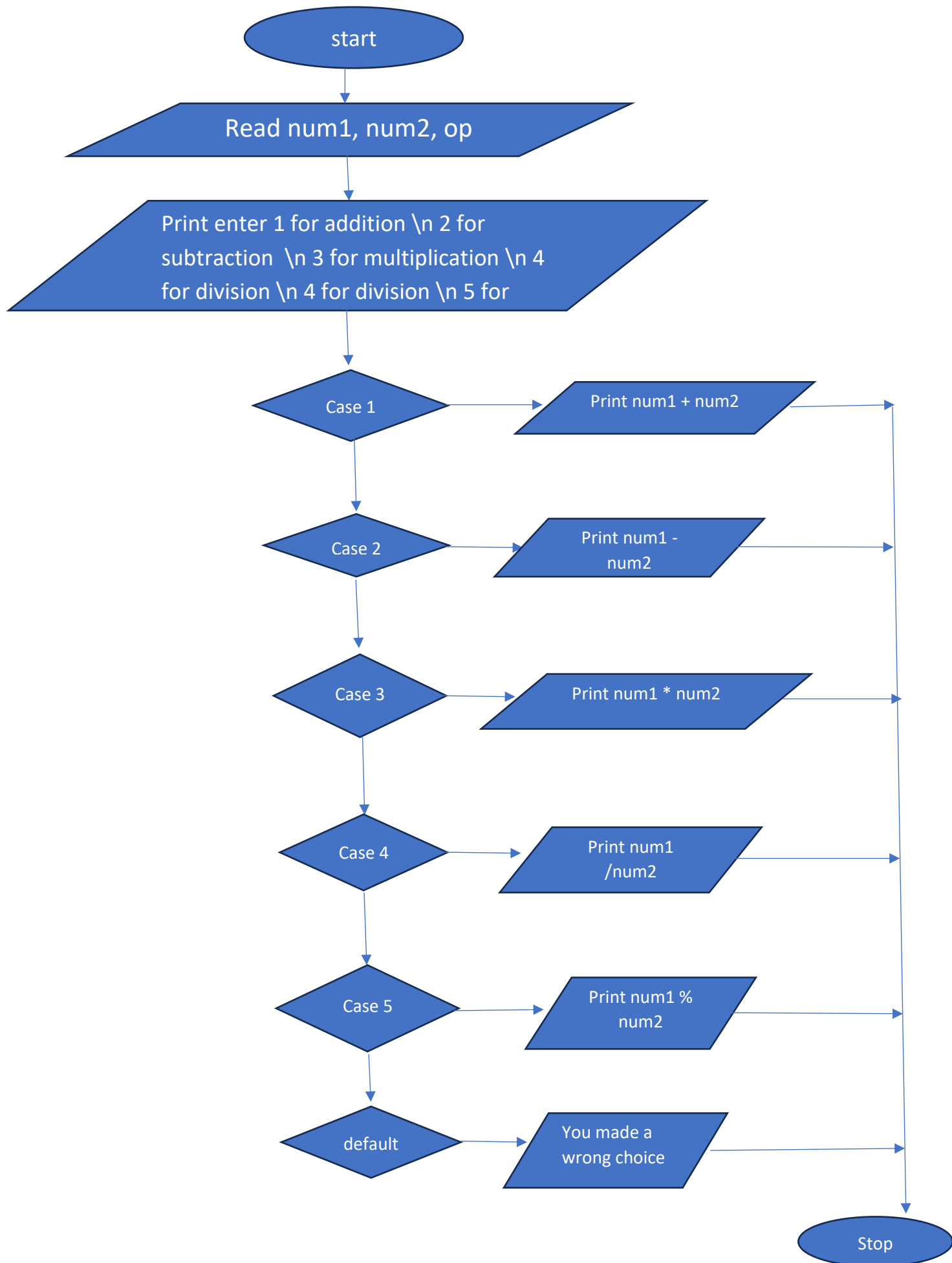
Print remainder when num 1
divide by num2 and break

Default

Print you made a wrong choice
And break

5. Stop

FLOWCHART



Q12:WRITE A PROGRAM TO PRINT SUM OF ALL NUMBERS UPTO A GIVEN NUMBER.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int num,i=0,sum=0;
clrscr();
printf("enter the number:");
scanf("%d",&num);
while(i<=num)
{
sum=sum+i;
i=i+1;
}
printf("sum of numbers upto %d is
%d",num,sum);
getch();
```

}

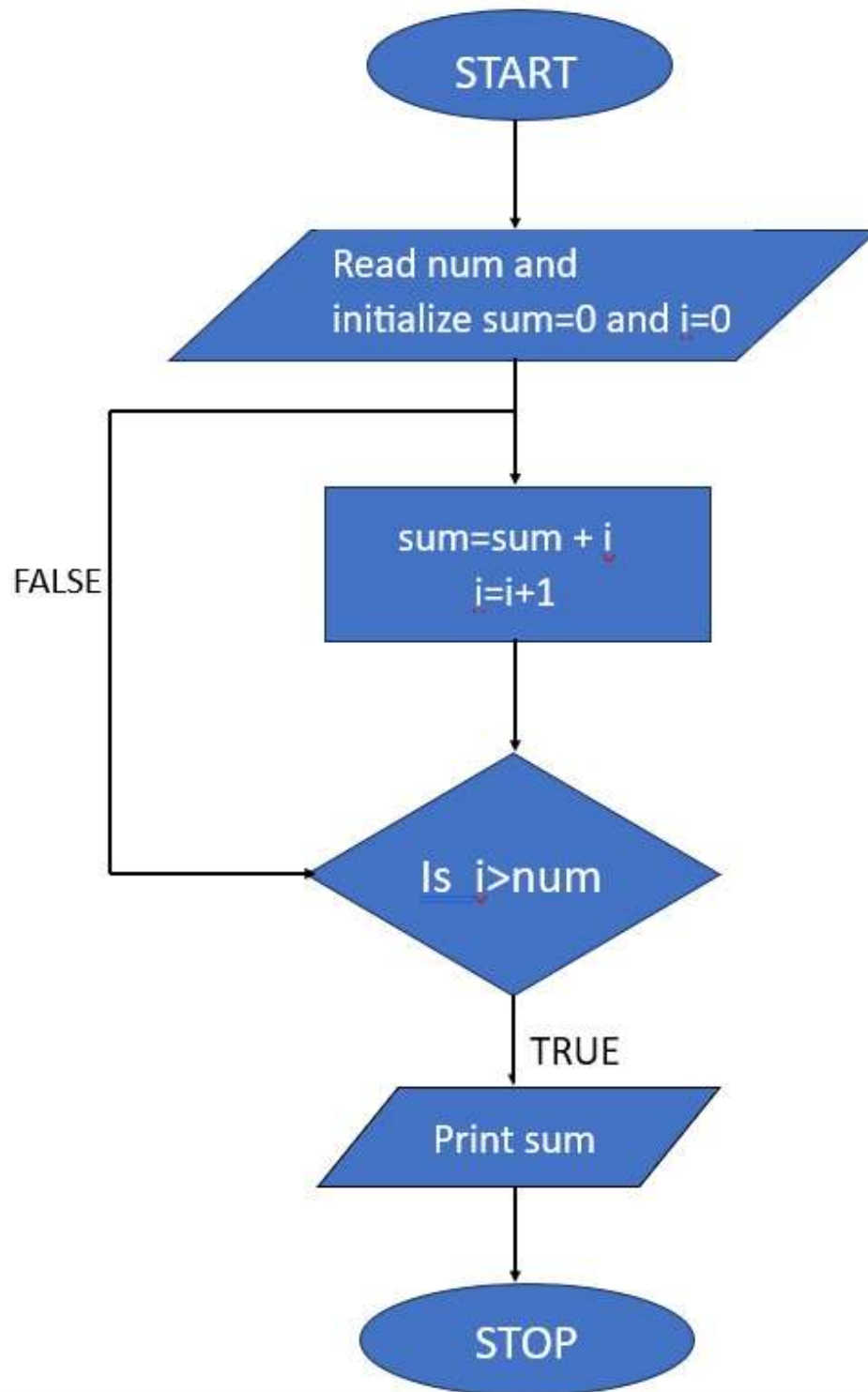
OUTPUT

```
C:\TURBOC3\BIN>TC
enter the number:7
sum of numbers upto 7 is 28_
```

ALGORITHM

- Start
- Read num,i,sum
- Initialise i=0,sum=0
- Sum=sum+i
 i=i+1
- Repeat steps 4 until i<=num
- Print sum of numbers
- Stop

FLOWCHART



Q13:WRITE A PROGRAM TO FIND FACTORIAL OF A NUMBER.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int num,fac=1,i=1;
clrscr();
printf("enter the number:");
scanf("%d",&num);
while(i<=num)
{
fac=fac*i;
i=i+1;
}
printf("the factorial of %d is %d",num,fac);
getch();
}
```

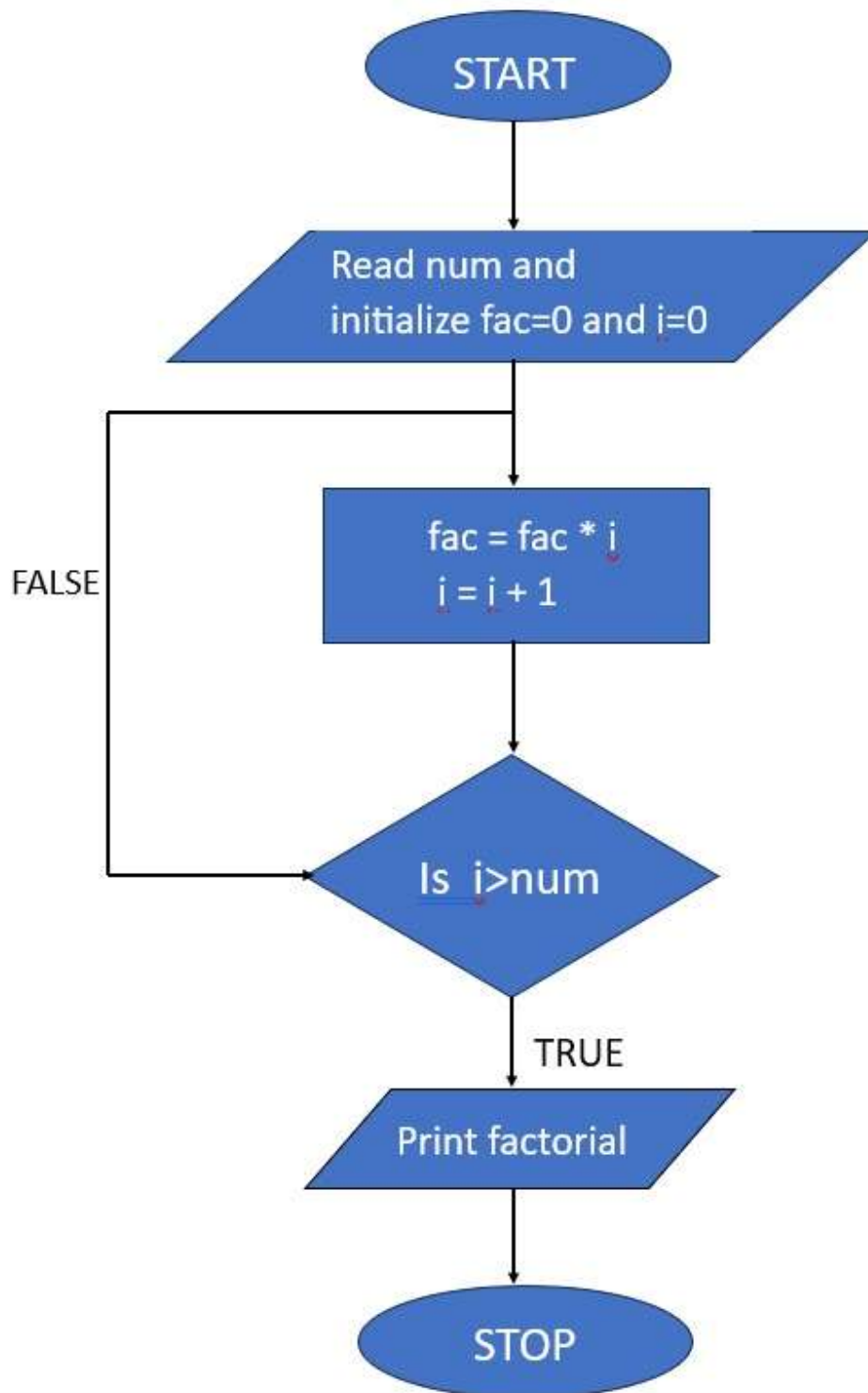
OUTPUT

```
enter the number:6  
the factorial of 6 is 720
```

ALGORITHM

- Start
- Read num and initialise fac=1 and i=1
- fac=fac*i
- i=i+1
- repeat steps 3 and 4 until i<=num
- print factorial of num
- stop

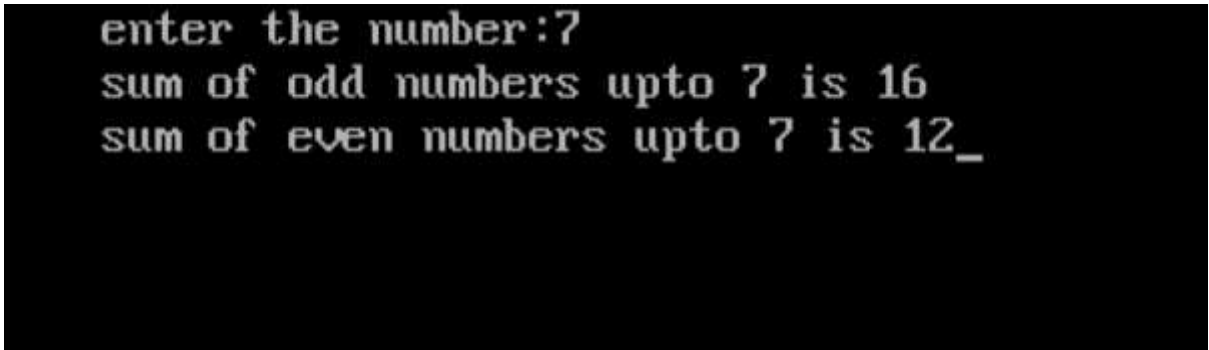
FLOWCHART



Q14:WRITE A PROGRAM TO PRINT SUM OF EVEN AND ODD NUMBERS FROM 1 TO N NUMBERS.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int num,sum_odd=0,sum_even=0,odd=-1,even=0;
clrscr();
printf("enter the number:");
scanf("%d",&num);
while(odd<=num)
{
sum_odd=sum_odd+odd;
odd=odd+2;
}
printf("sum of odd numbers upto %d is
%d\n",num,odd);
while(even<=num)
```

```
{  
    sum_even=sum_even+even;  
    even=even+2;  
}  
  
printf("sum of even numbers upto %d is  
%d",num,even);  
getch();  
}
```



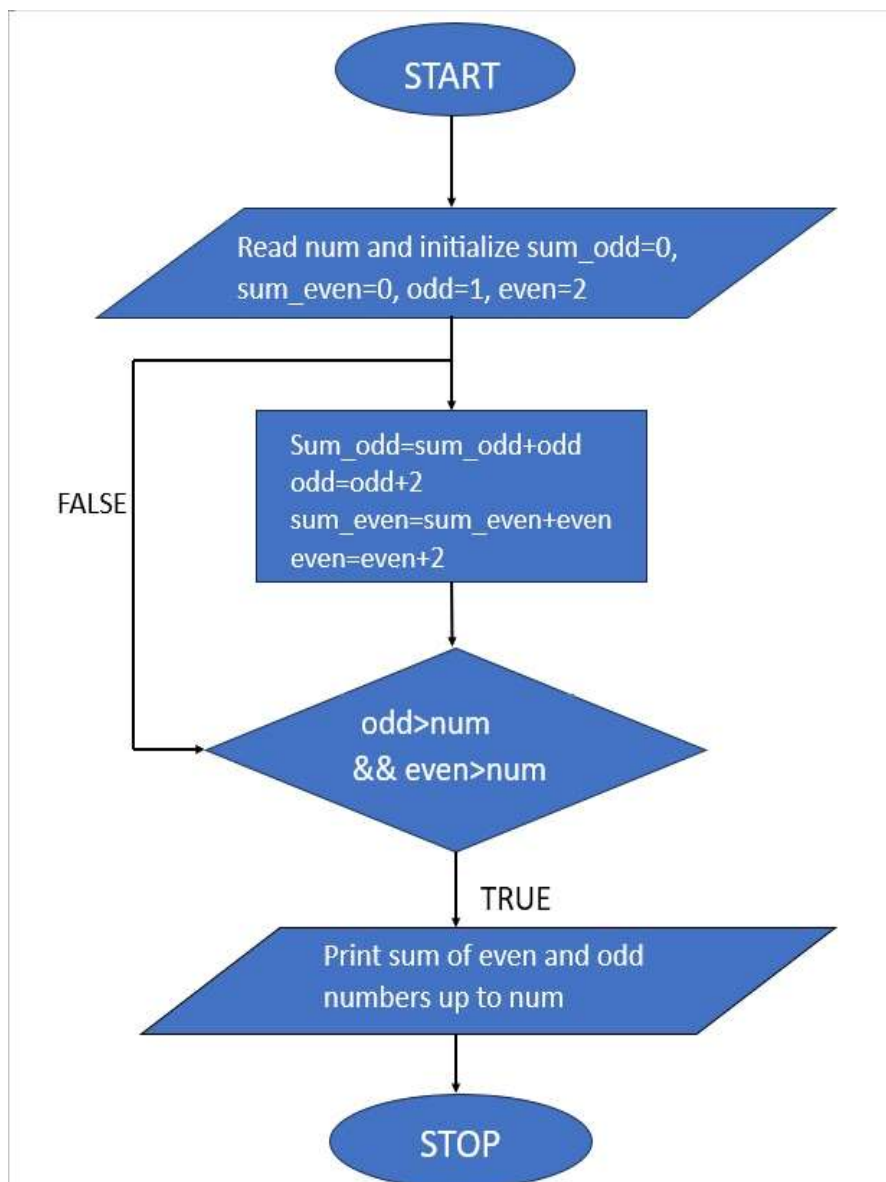
```
enter the number:7  
sum of odd numbers upto 7 is 16  
sum of even numbers upto 7 is 12_
```

ALGORITHM

- start
- read num and initialise
 sum_odd=0,sum_even=0,odd=1,even=2
- sum_odd=sum_odd+odd
 odd=odd+2

- $\text{sum_even} = \text{sum_even} + \text{even}$
 $\text{even} = \text{even} + 2$
- repeat steps 3 and 4 until $\text{odd} \leq \text{num}$ and $\text{even} \leq \text{num}$
- print sum of odd and even numbers
- stop

FLOWCHART

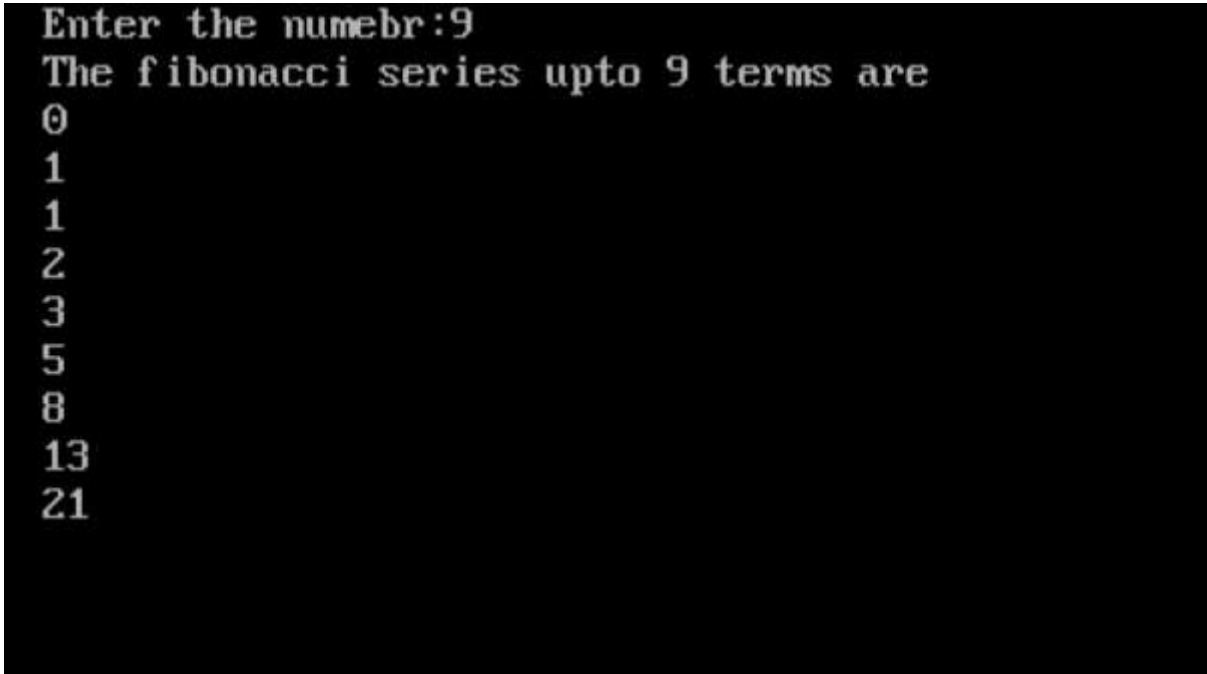


Q15:WRITE A PROGRAM TO PRINT FIBONACCI SERIES.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int num,i=0,f1=0,f2=1,f3=0;
clrscr();
printf("Enter the numebr:");
scanf("%d",&num);
printf("The fibonacci series upto %d terms
are\n",num);
printf("%d\n",f1);
printf("%d\n",f2);
for(i=0;i<num-2;i++)
{
f3=f1+f2;
printf("%d\n",f3);
f1=f2;
```

```
f2=f3;  
}  
getch();  
}
```

OUTPUT



```
Enter the numebr:9  
The fibonacci series upto 9 terms are  
0  
1  
1  
2  
3  
5  
8  
13  
21
```

ALGORITHM

- start
- read num and initialise i=0,f1=0,f2=1 and f3=0
- print the Fibonacci series upto n terms is
 - print f1
 - print f2
- $f3=f1+f2$

print f3

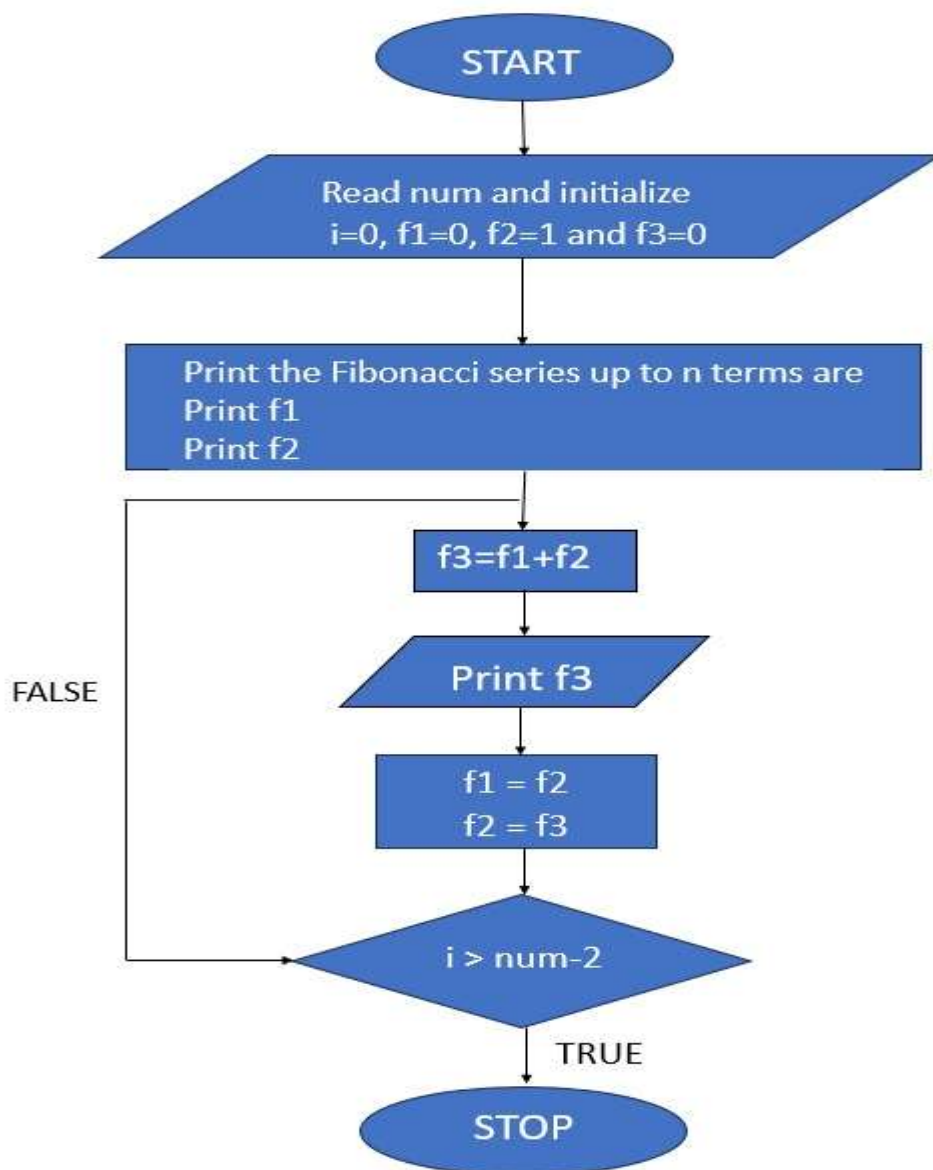
f1=f2

f2=f3

➤ repeat step 4 until $i < \text{num}-2$

➤ stop

FLOWCHART

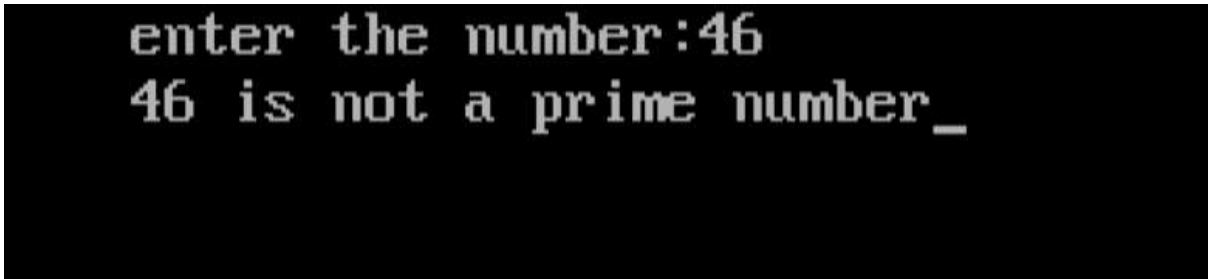


Q16:WRITE A PROGRAM TO CHECK WHETHER THE ENTERED NUMBER IS PRIME OR NOT.

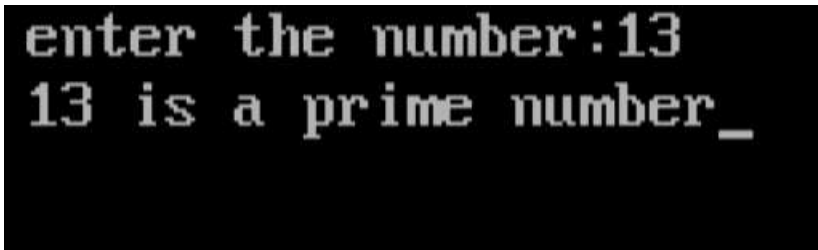
```
#include<stdio.h>
#include<conio.h>
void main()
{
int num,count=0,f=1,i=1;
clrscr();
printf("enter the number:");
scanf("%d",&num);
for(i=1;i<=num;i++)
{
if(num%i==0)
{
count=count+1;
}
}
if(count==2)
```

```
{  
    printf("%d is a prime number",num);  
}  
else  
{  
    printf("%d is not a prime number",num);  
}  
getch();  
}
```

OUTPUT



```
enter the number:46  
46 is not a prime number_
```



```
enter the number:13  
13 is a prime number_
```

ALGORITHM

1 start

2 read num and initialise count=0 and i=1

3 if $\text{num} \% i == 0$

 Count =count+1

4 repeat step 3 until $i \leq \text{num}$

5 if count ==?

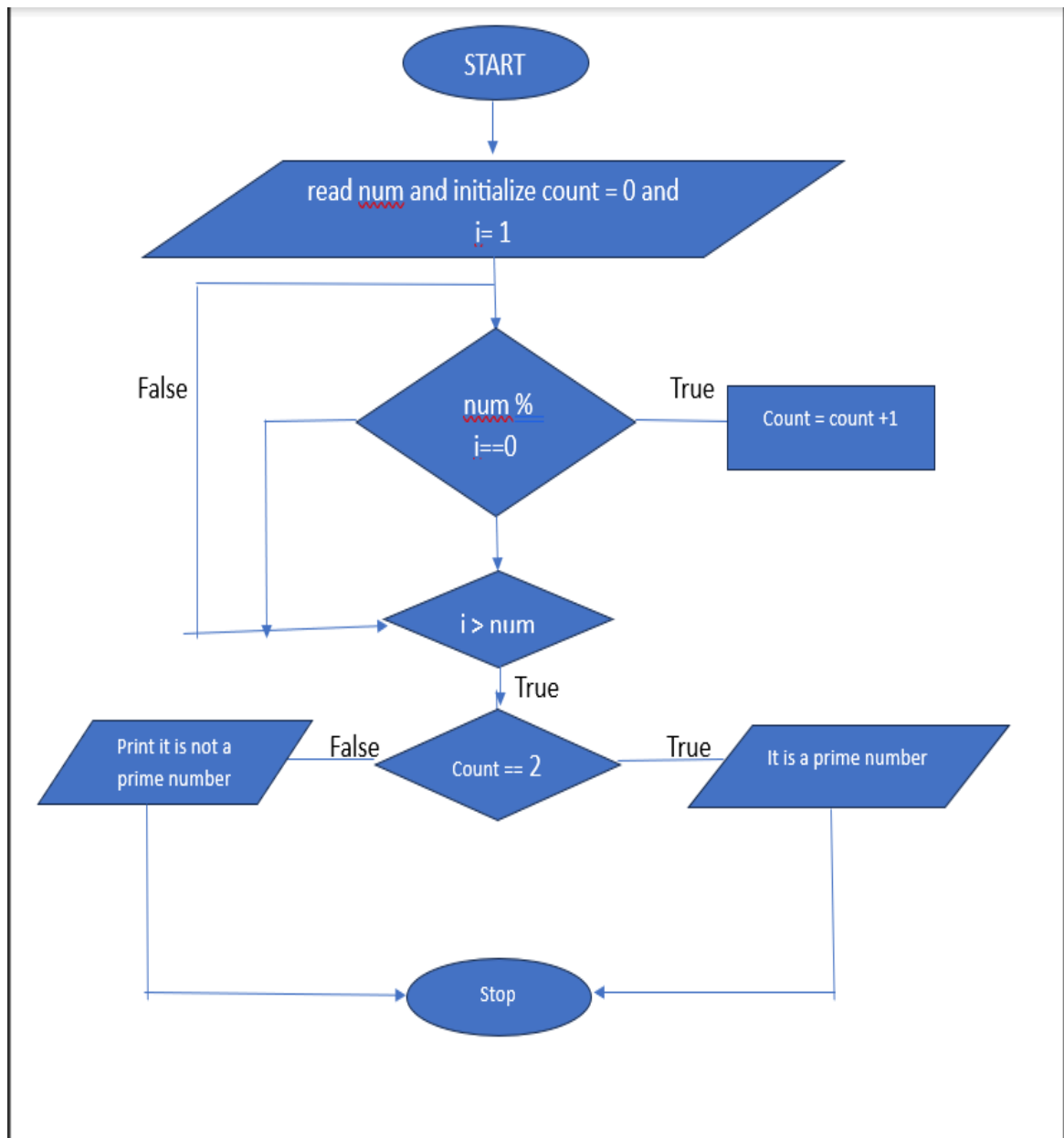
 Print it is a prime number

 else

 print it is not a prime number

6 stop

FLOWCHART

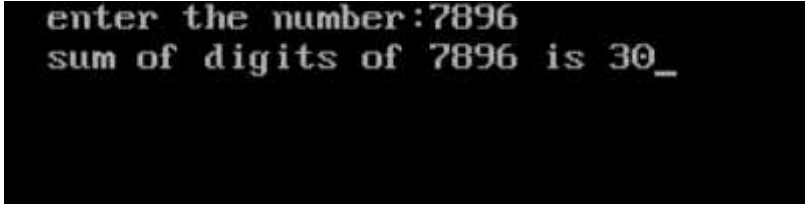


Q17:WRITE A PROGRAM TO FIND THE SUM OF DIGITS OF THE ENTERED NUMBER.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,num,rem=0,sum=0;
clrscr();
printf("enter the number:");
scanf("%d",&num);
n=num;
while(num>0)
{
rem=num%10;
sum=rem+sum;
num=num/10;
}
printf("sum of digits of %d is %d",n,sum);
```

```
getch();  
}
```

OUTPUT

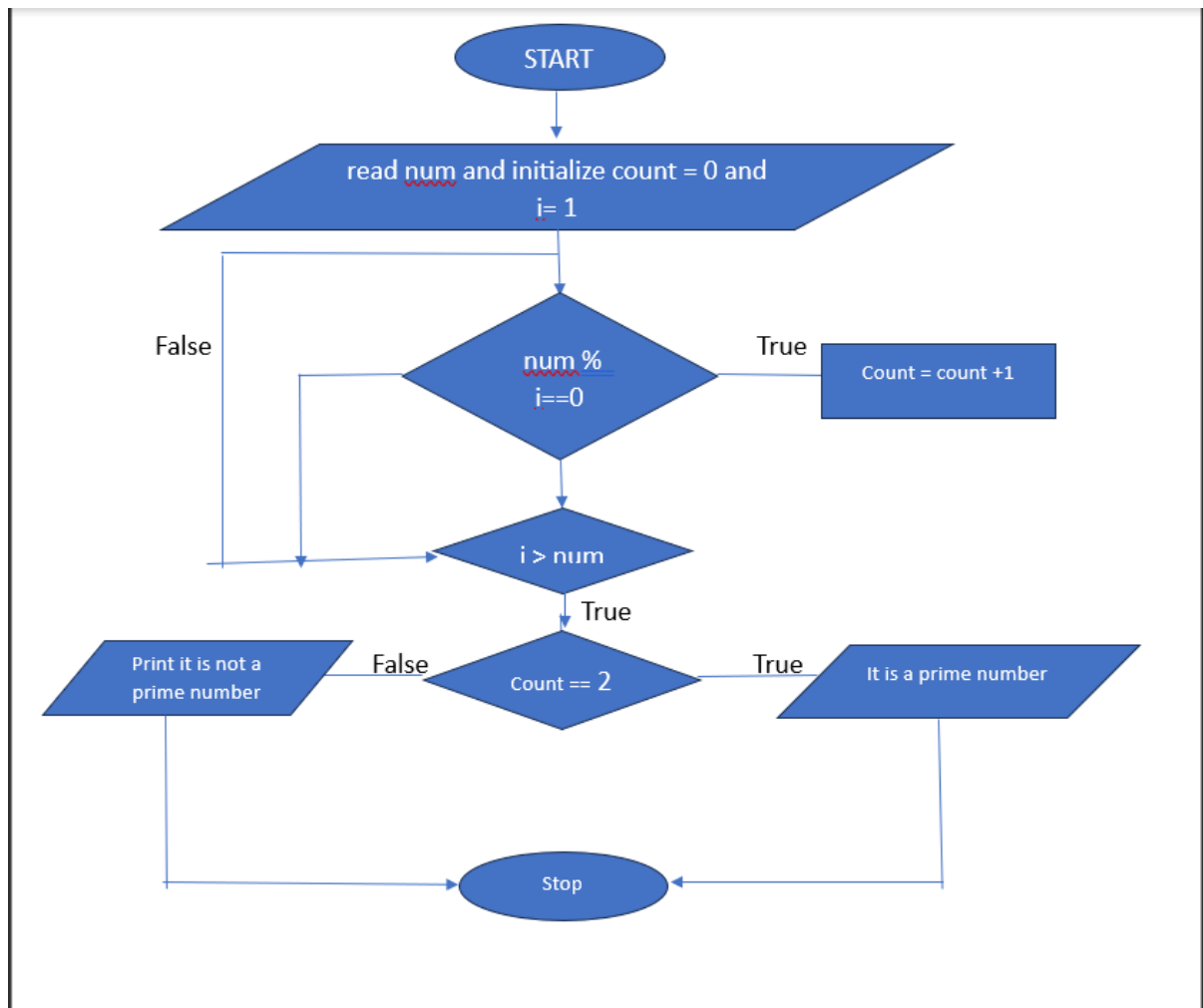


```
enter the number:7896  
sum of digits of 7896 is 30_
```

ALGORITHM

- 1 start
- 2 read n, num and initialise rem=0 , sum=0
- 3 rem= num% 10
 Sum = rem+ sum
 num = num/10
- 4 repeat step 3 until num>0
- 5 print the sum of digits
- 6 stop

FLOWCHART

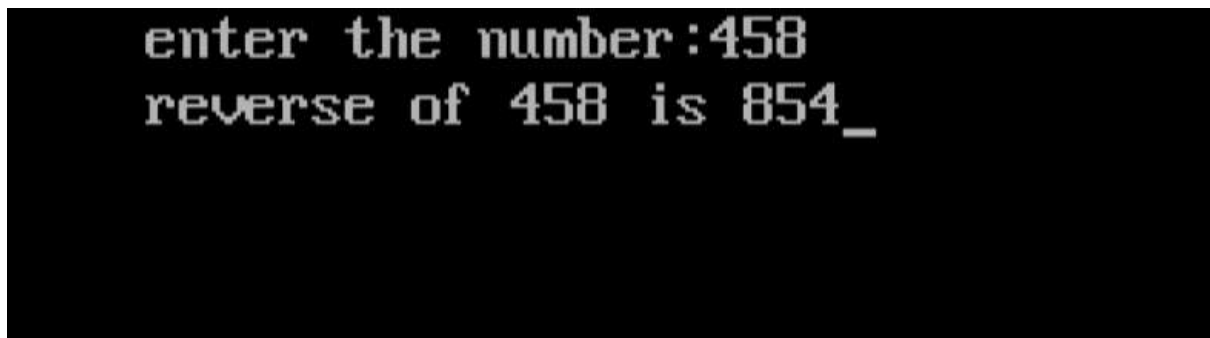


Q18:WRITE A PROGRAM TO FIND REVERSE OF A NUMBER.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,num,rem=0,rev=0;
clrscr();
printf("enter the number:");
scanf("%d",&num);
n=num;
while(num>0)
{
rem=num%10;
rev=rem+rev*10;
num=num/10;
}
```

```
printf("reverse of %d is %d",n,rev);  
getch();  
}
```

OUTPUT

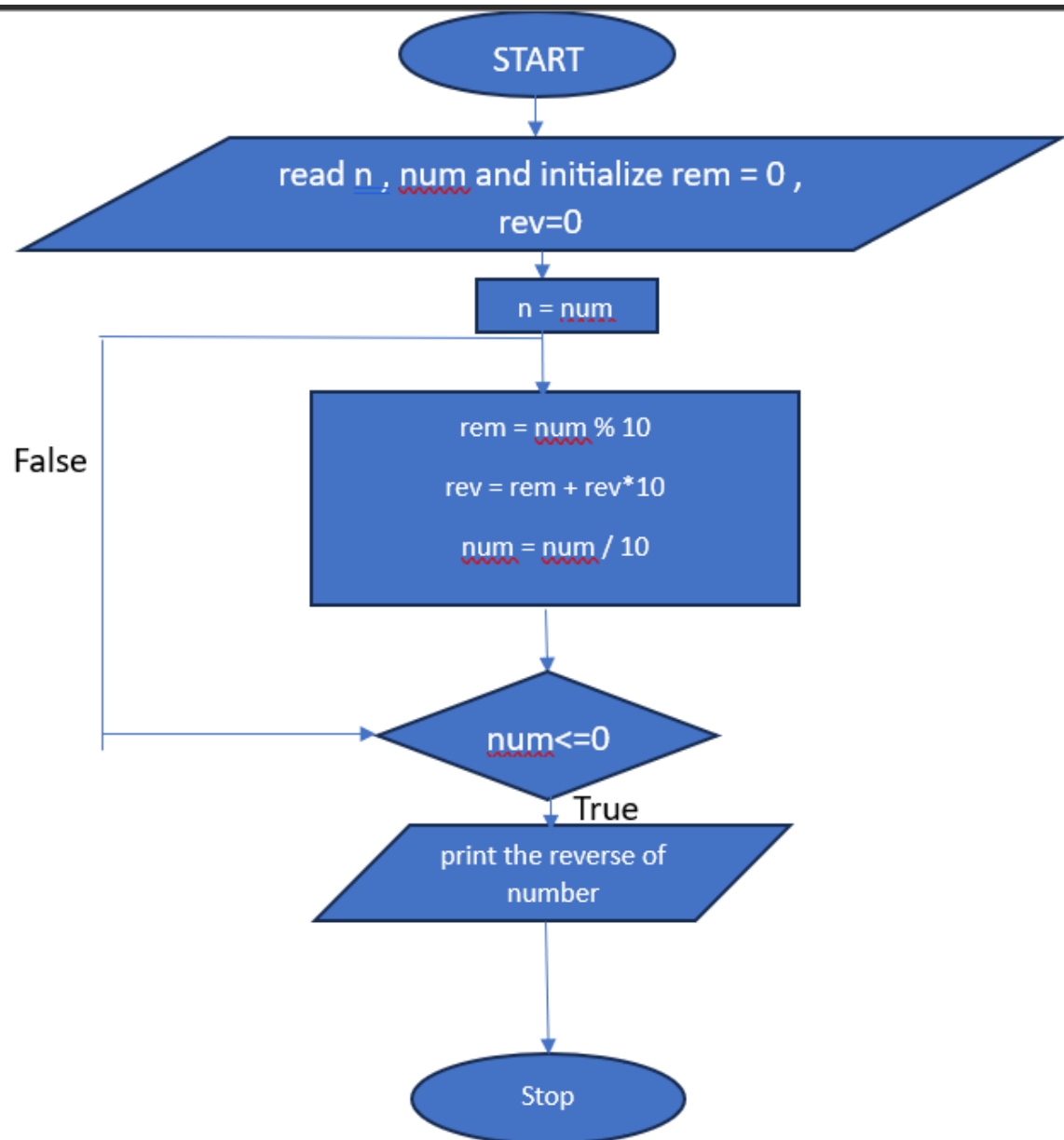


```
enter the number:458  
reverse of 458 is 854_
```

ALGORITHM

- 1 . start
2. read n, num and initialise rem=0, rev=0
- 3 . n=num
4. rem=num%10
 rev = rem +rev*10
 num = num/10 repeat step 4 until num>0
- 5 . print the reverse of number
- 6 . stop

FLOWCHART



Q19:WRITE A PROGRAM TO PRINT ARMSTRONG NUMBERS FROM 1 TO 1000.

```
#include <stdio.h>
#include <math.h>
void main()
{
    printf("All Armstrong numbers between 1 and 1000
are:\n");
    for(int n=1;n<=1000;n++)
    {
        int num = n,rem,sum=0;
        if(n<=9)
        {
            printf("%d\n",n);
        }
        else
        {
            int digit = (int) log10(num) + 1; //To count
number of digits
```

```
//Calculating sum of power of digits of a number
while(num > 0)
{
    rem = num % 10; //To find last digit of the
number
    sum = sum + pow(rem,digit);
    num = num / 10;
}

if (sum == n)
{
    printf("%d ", n);.
}
}
}
```

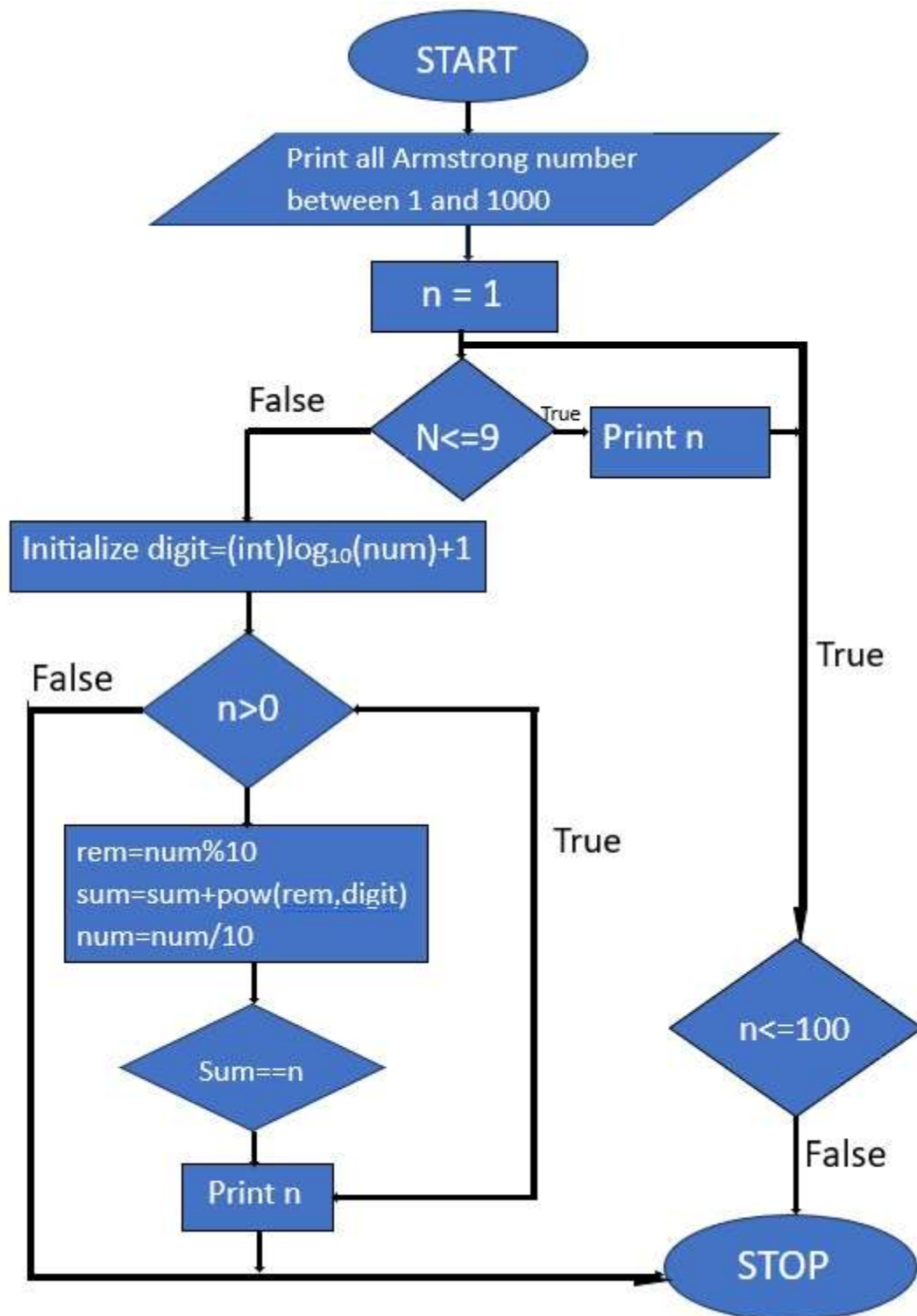
OUTPUT

```
Output
/tmp/WQPE57oBUC.o
All Armstrong numbers between 1 and 1000 are:
1
2
3
4
5
6
7
8
9
153
370
371
407
|
```

ALGORITHM

- start
- print all Armstrong numbers between 1 and 1000
- read $n=1$
- if $n \leq 9$
 - print n
- else
 - intialise $digit = \text{intlog}_{10}(num+1)$
 - $rem = num \% 10$
 - $sum = sum + \text{pow}(rem, digit)$
 - $num = num / 10$
 - repeat previois step until $num > 0$
 - repeat previous step until $n < 1000$
 - stop

FLOWCHART



**Q20:WRITE A PROGRAM TO CONVERT DECIMAL
NUMBER TO BINARY NUMBER AND VICE VERSA.**

```
#include<stdio.h>

void main()
{
    int num,n,rem,dec,bin,base=1,ch;
    printf("Enter 1 to perform binary conversion \n
2 for decimal conversion:");
    scanf("%d",&ch);
    if(ch==1)
    {
        printf("Enter the decimal number:");
        scanf("%d",&num);
        n=num;
        while(num>0)
        {
            rem=num%2;
            bin=bin+rem*base;
```

```
        num=num/2;
        base=base*10;
    }
    printf("Binary number of %d is %d",n,bin);
}
else if(ch==2)
{
    printf("Enter the binary number:");
    scanf("%d",&num);
    n=num;
    while(num>0)
    {
        rem=num%10;
        dec=dec+rem*base;
        num=num/10;
        base=base*2;
    }
    printf("Decimal number of %d is %d",n,dec);
}
```

```
}
```

OUTPUT

Output

```
/tmp/xZVohQwWym.o  
Enter 1 to perfrom binary conversion  
2 for decimal conversion:1  
Enter the decimal number:8  
Binary number of 8 is 1000|
```

Output

```
/tmp/xZVohQwWym.o  
Enter 1 to perfrom binary conversion  
2 for decimal conversion:2  
Enter the binary number:1000  
Decimal number of 1000 is 8
```

ALGORITHM

- start
- read num,n,rem,dec,bin,ch and
initialise base =1
- print enter 1 to perform binary
conversion and 2 for decimal conversion
- if ch=1
enter the decimal number
n=num
- rem=num%2
bin=bin+rem*base
num=num/2
base=base*10
- repeat previous step until num>0
print binary remainder
- else if ch==2
enter the binary number
n=num
- rem=num%10

$\text{dec} = \text{dec} + \text{rem} * \text{base}$

$\text{num} = \text{num} / 10$

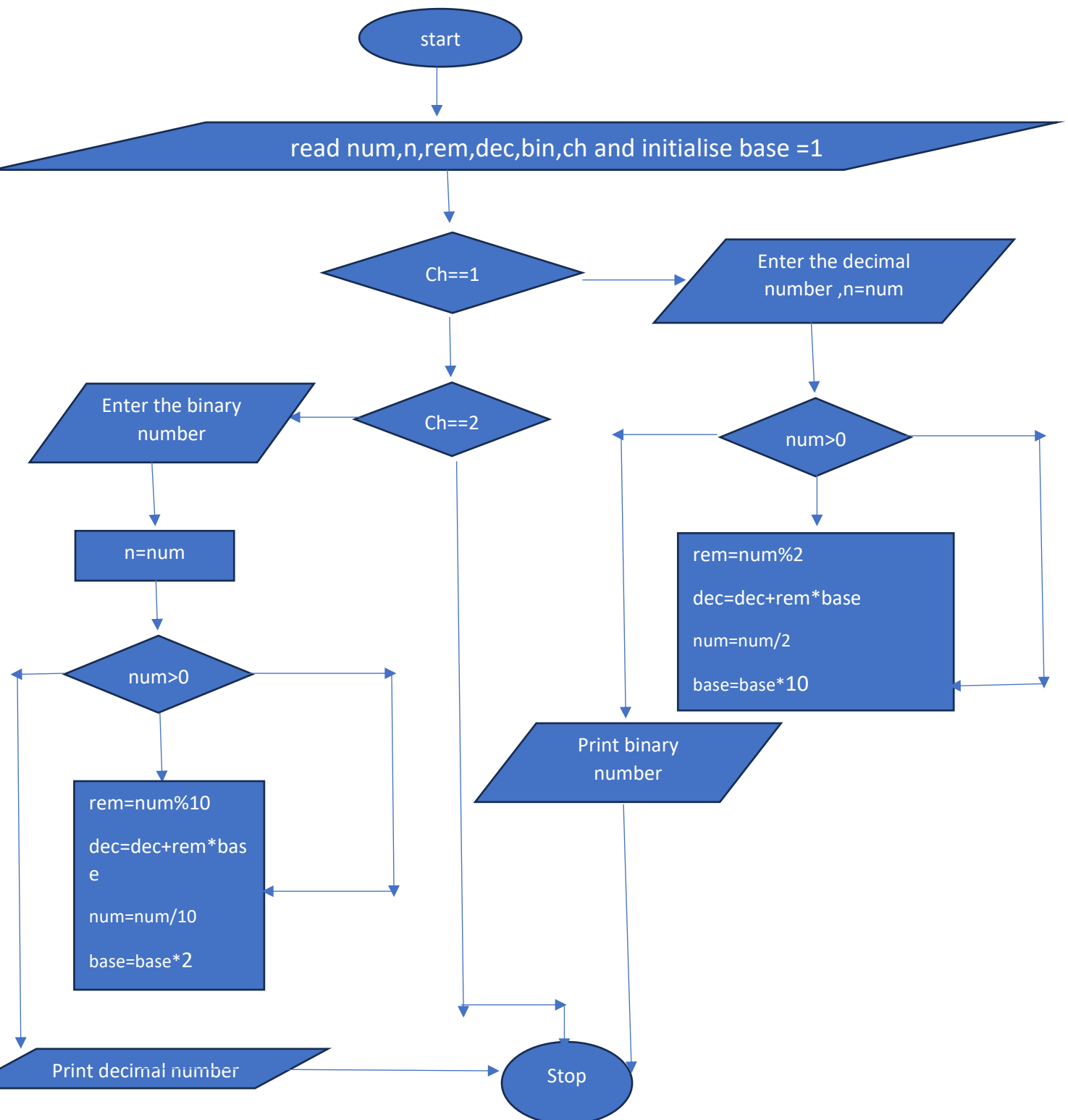
$\text{base} = \text{base} * 2$

➤ repeat previous step until $\text{num} > 0$

print the decimal number

➤ stop

FLOWCHART

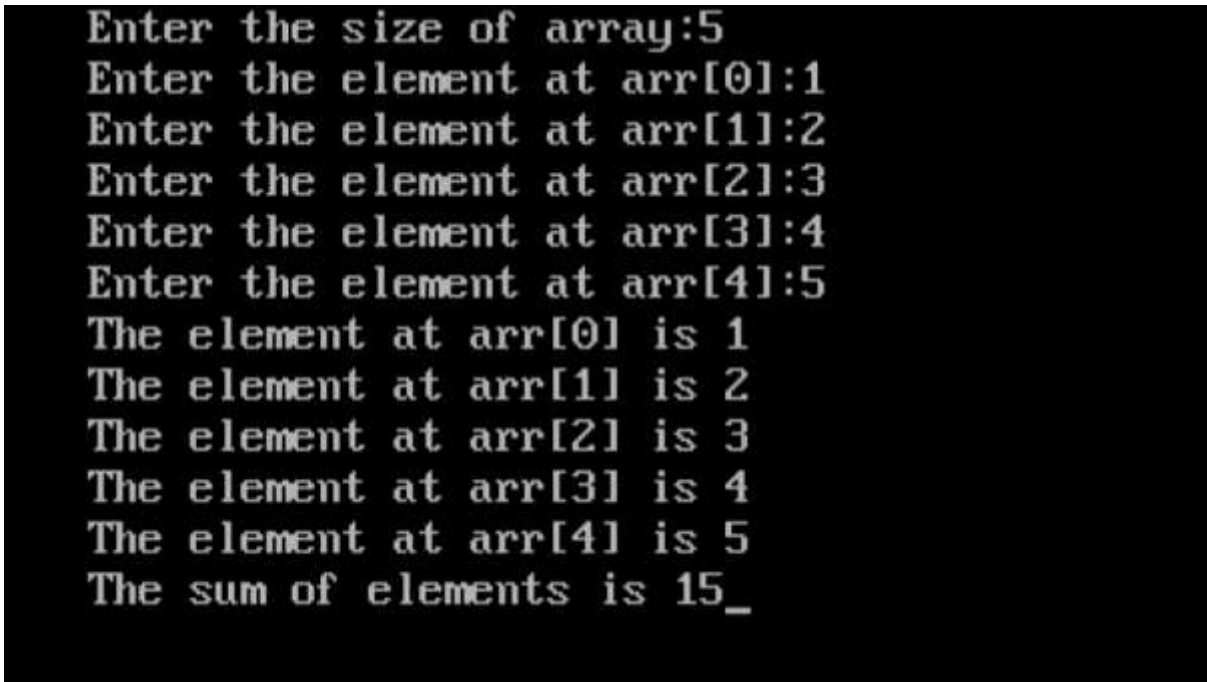


**Q21:WRITE A PROGRAM THAT SIMPLY TAKES
ELEMENT OF THE ARRAY FROM THE USER AND FINDS
THE SUM OF THESE ELEMENTS.**

```
#include<stdio.h>
#include<conio.h>
void main()
{
int arr[50],i,size,sum=0;
clrscr();
printf("Enter the size of array:");
scanf("%d",&size);
for(i=0;i<size;i++)
{
printf("Enter the element at arr[%d]:",i);
scanf("%d",&arr[i]);
}
for(i=0;i<size;i++)
```

```
{  
    sum=sum+arr[i];  
    printf("The element at arr[%d] is  
    %d\n",i,arr[i]);  
}  
printf("The sum of elements is %d",sum);  
getch();  
}
```

OUTPUT

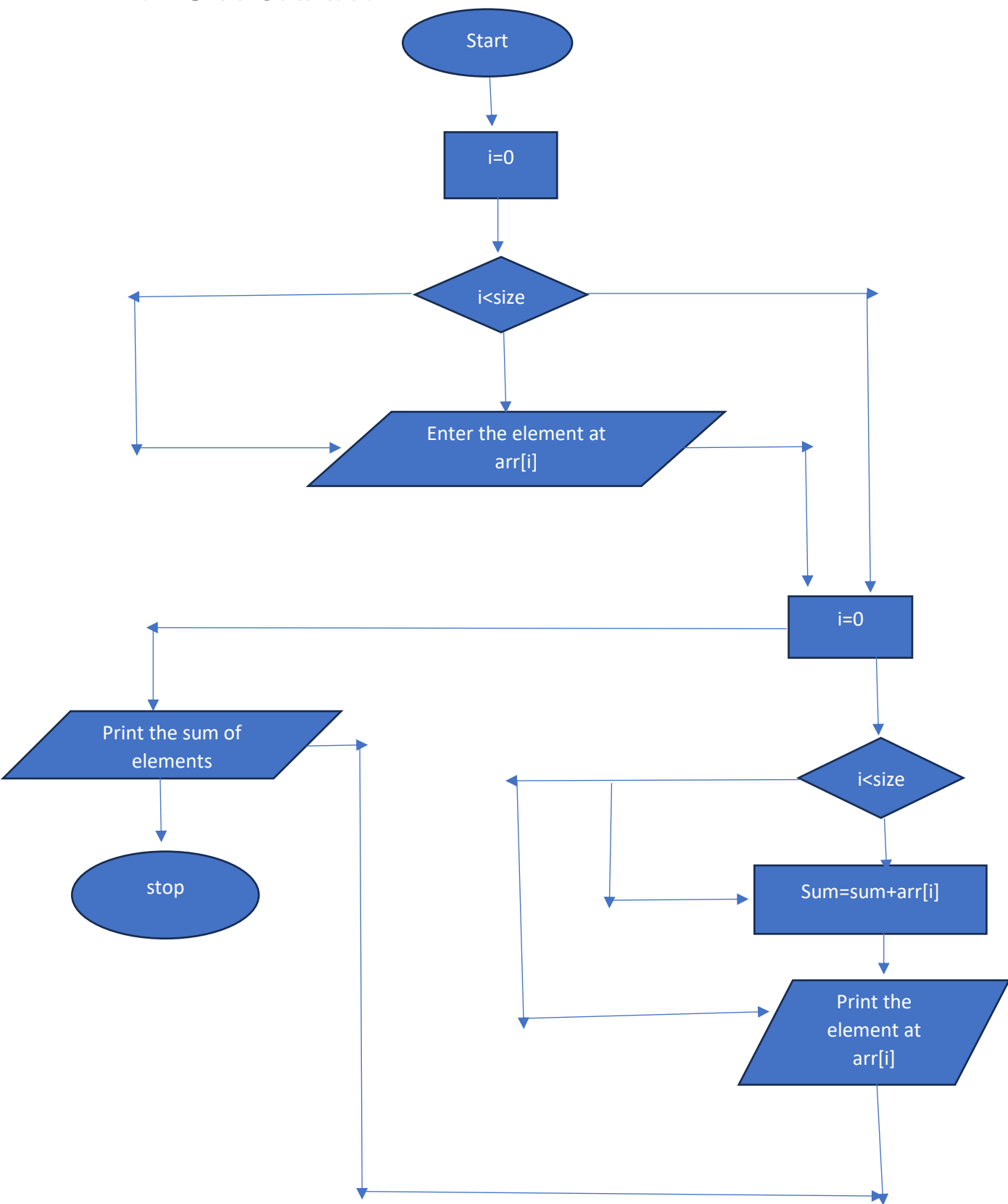
A screenshot of a terminal window with a black background and white text. The text shows the execution of a C program. It starts with a prompt to enter the size of the array (5), followed by five prompts to enter elements at indices 0 through 4 (values 1 through 5). Then, it displays the value of each element at its respective index, and finally, it shows the sum of all elements as 15.

```
Enter the size of array:5  
Enter the element at arr[0]:1  
Enter the element at arr[1]:2  
Enter the element at arr[2]:3  
Enter the element at arr[3]:4  
Enter the element at arr[4]:5  
The element at arr[0] is 1  
The element at arr[1] is 2  
The element at arr[2] is 3  
The element at arr[3] is 4  
The element at arr[4] is 5  
The sum of elements is 15_
```

ALGORITHM

- start
- read arr[50],i,size and initialise sum=0
- initialise i=0
- enter the element at arr[i],print the element ,sum=sum+arr[i]
- increment i by 1
- repeat step 4 and 5 until i<size
- print sum
- stop

FLOWCHART



Q22:WRITE A PROGRAM THAT TAKES INPUT OF TWO ARRAYS AND SAVES SUM OF CORRESPONDING ELEMENTS OF THESE ARRAYS IN A THIRD ARRAY AND PRINTS THEM.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[50],b[50],c[50],i,size,sum=0;
clrscr();
printf("Enter the size of array:");
scanf("%d",&size);
printf("The elements of first array are:");
for(i=0;i<size;i++)
{
printf("Enter the element at arr[%d]:",i);
scanf("%d",&a[i]);
}
```

```
printf("The elements of second array are:\n");
for(i=0;i<size;i++)
{
    printf("Enter the element at arr[%d]:",i);
    scanf("%d",&b[i]);
}
for(i=0;i<size;i++)
{
    c[i]=a[i]+b[i];
}
printf("The sum of elements of array1 and array2
is");
for(i=0;i<size;i++)
{
    printf("%d\n",c[i]);
}
getch();
}
```

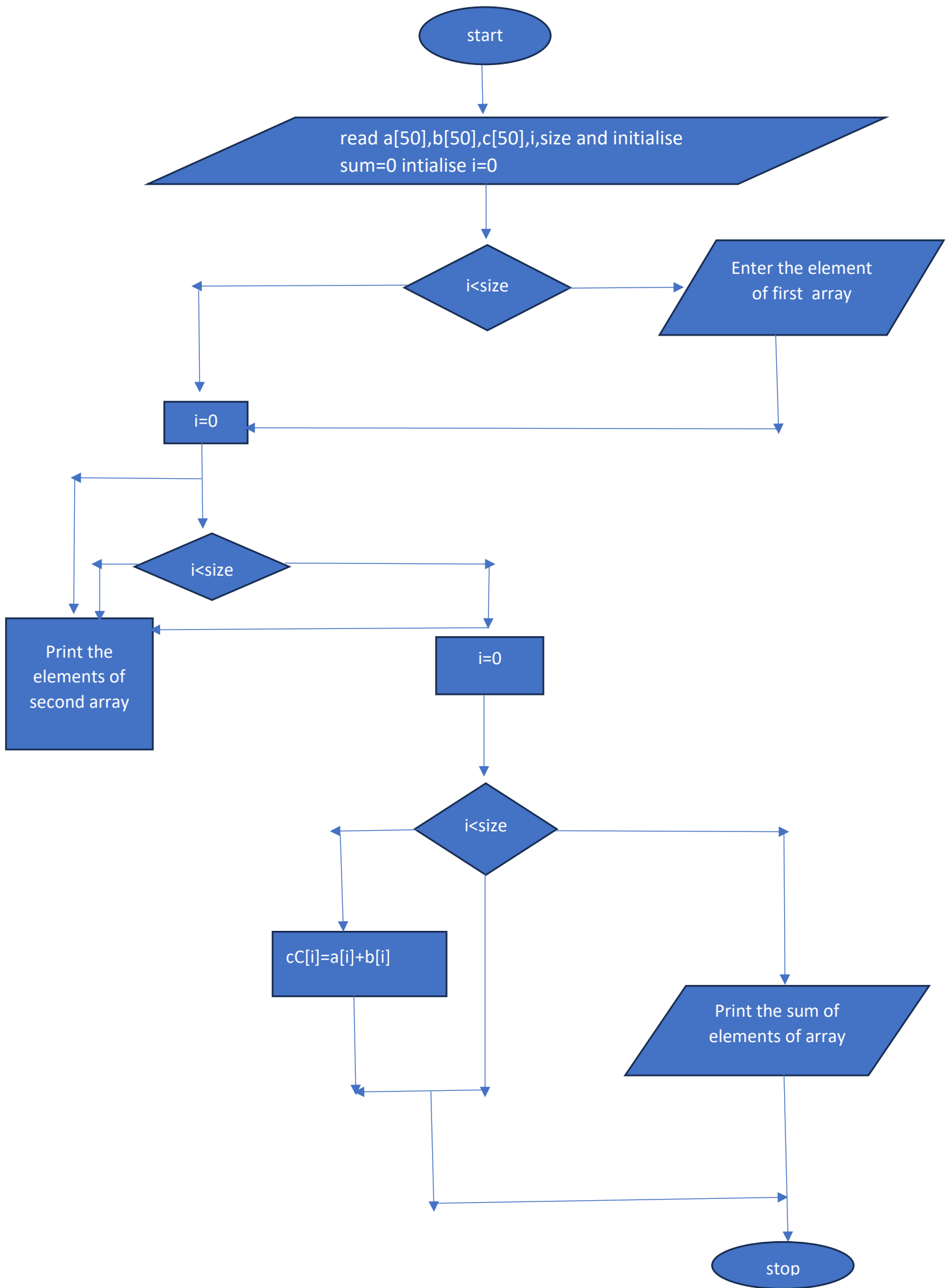
OUTPUT

```
Enter the size of array:4
The elements of first array are:Enter the element at arr[0]:1
Enter the element at arr[1]:2
Enter the element at arr[2]:3
Enter the element at arr[3]:4
The elements of second array are:
Enter the element at arr[0]:1
Enter the element at arr[1]:2
Enter the element at arr[2]:3
Enter the element at arr[3]:4
The sum of elements of array1 and array2 is2
4
6
8
-
```

ALGORITHM

- start
- read a[50],b[50],c[50],i,size and initialise
sum=0
- initialise i=0
- enter the element at arr[i] for array 1 and
array 2 ,c[i]=a[i]+b[i] and print the element of
array 1 and array 2
- increment i by 1
- repeat step 4 and 5 until i<size
- stop

FLOWCHART



Q23:WRITE A PROGRAM TO FIND MAXIMUM AND MINIMUM ELEMENT OF ARRAY.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[50],size,i,max=0,min=0;
printf("Enter the size of array");
scanf("%d",&size);
for(i=0;i<size;i++)
{
printf("Enter the element at a[%d]",i);
scanf("%d",&a[i]);
}
max=a[0];
min=a[0];
for(i=0;i<size;i++)
{
if(max<a[i])
```

```

{
    max=a[i];
}
if(min>a[i])
{
    min=a[i];
}
}

printf("The maximum and minimum values of
given array is %d and %d",max,min);
getch();
}

```

OUTPUT

```

C:\TURBOC3\BIN>TC
Enter the size of array5
Enter the element at a[0]1
Enter the element at a[1]9
Enter the element at a[2]-8
Enter the element at a[3]67
Enter the element at a[4]4
The maximum and minimum values of given array is 67 and -8_

```

ALGORITHM

➤ START

➤ Read $a[50]$, size, i and initialise $\text{max}=0$, $\text{min}=0$

➤ Initialise $i=0$

➤ Enter the element at $a[i]$, increment i by 1

➤ Repeat step 4 until $i < \text{size}$

➤ $\text{Max}=a[0]$ and $\text{min}=a[0]$

➤ If ($\text{max} < a[i]$)

$\text{Max}=a[i]$

If ($\text{min} < a[i]$)

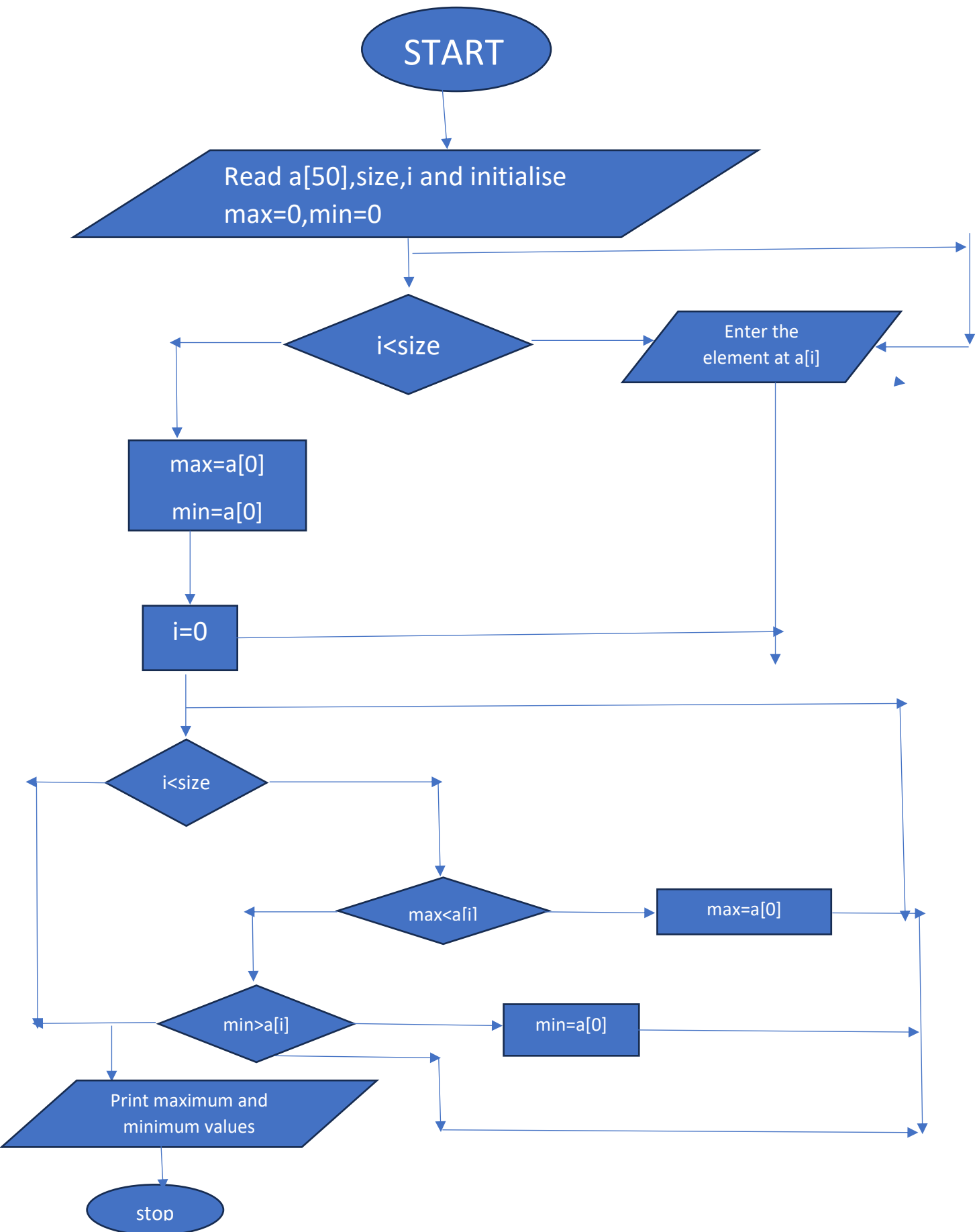
$\text{Min}=a[i]$

➤ Repeat step 7 until $i < \text{size}$

➤ Print the maximum and minimum values of
given array

➤ Stop

FLOWCHART



24. WAP TO SEARCH AN ELEMENT IN AN ARRAY USING LINEAR SEARCH.

```
#include<stdio.h>

int search(int array[],int n,int x)
{
    for(int i=0;i<n;i++)
    {
        if(array[i]==x)
        {
            return i;
        }
    }
    return -1;
}

void main()
{
    int array[20];
    int n;
    int x;
    printf("Enter the number of elements:");
```

```
scanf("%d",&n);
for(int i=0;i<n;i++)
{
    printf("Enter Element at array[%d]:",i);
    scanf("%d",&array[i]);
}
printf("Enter the element to be searched:");
scanf("%d",&x);
int result=search(array,n,x);
if(result==-1)
{
    printf("Element not found");
}
else
{
    printf("%d is at index %d",x,result);
}
}
```

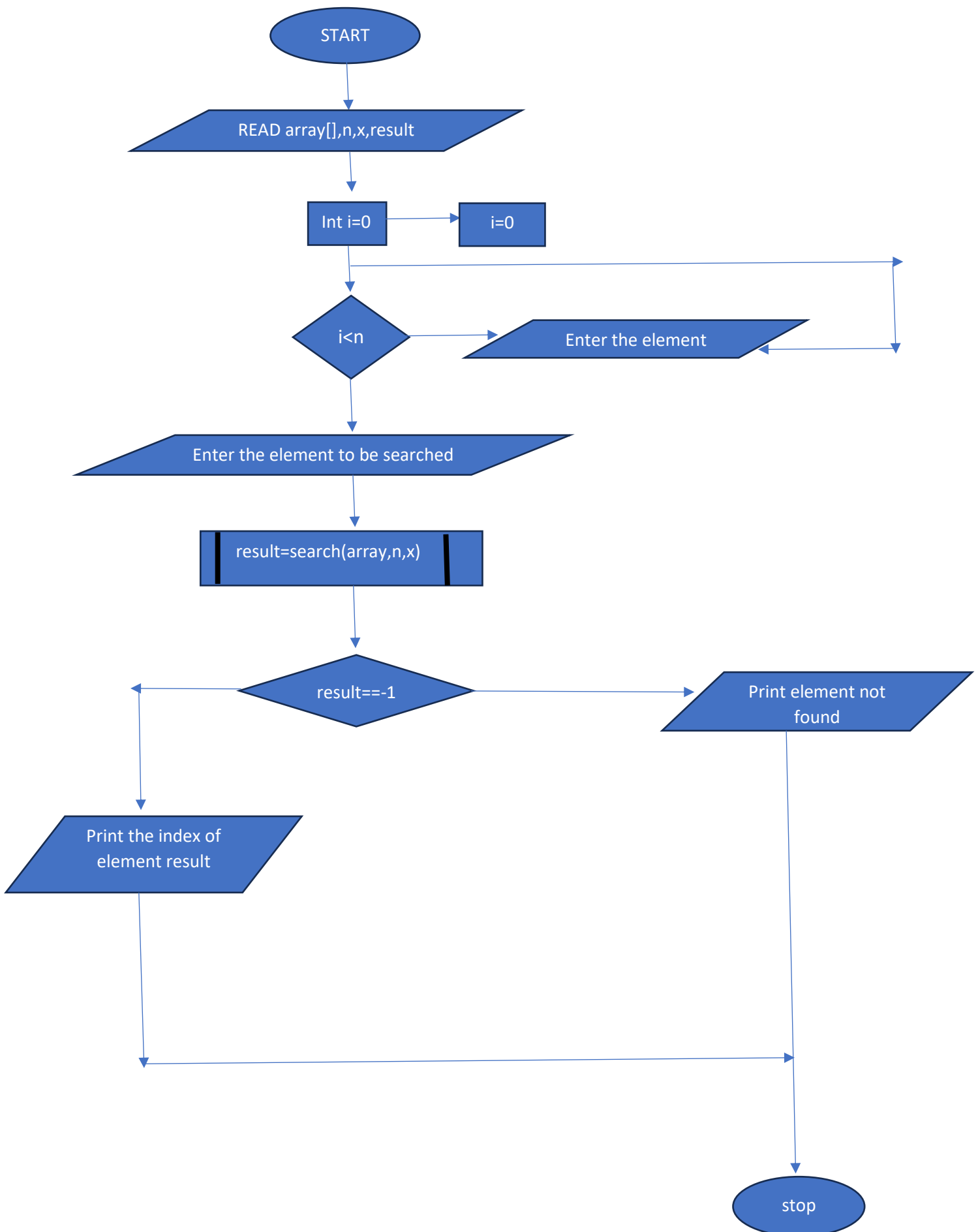
OUTPUT

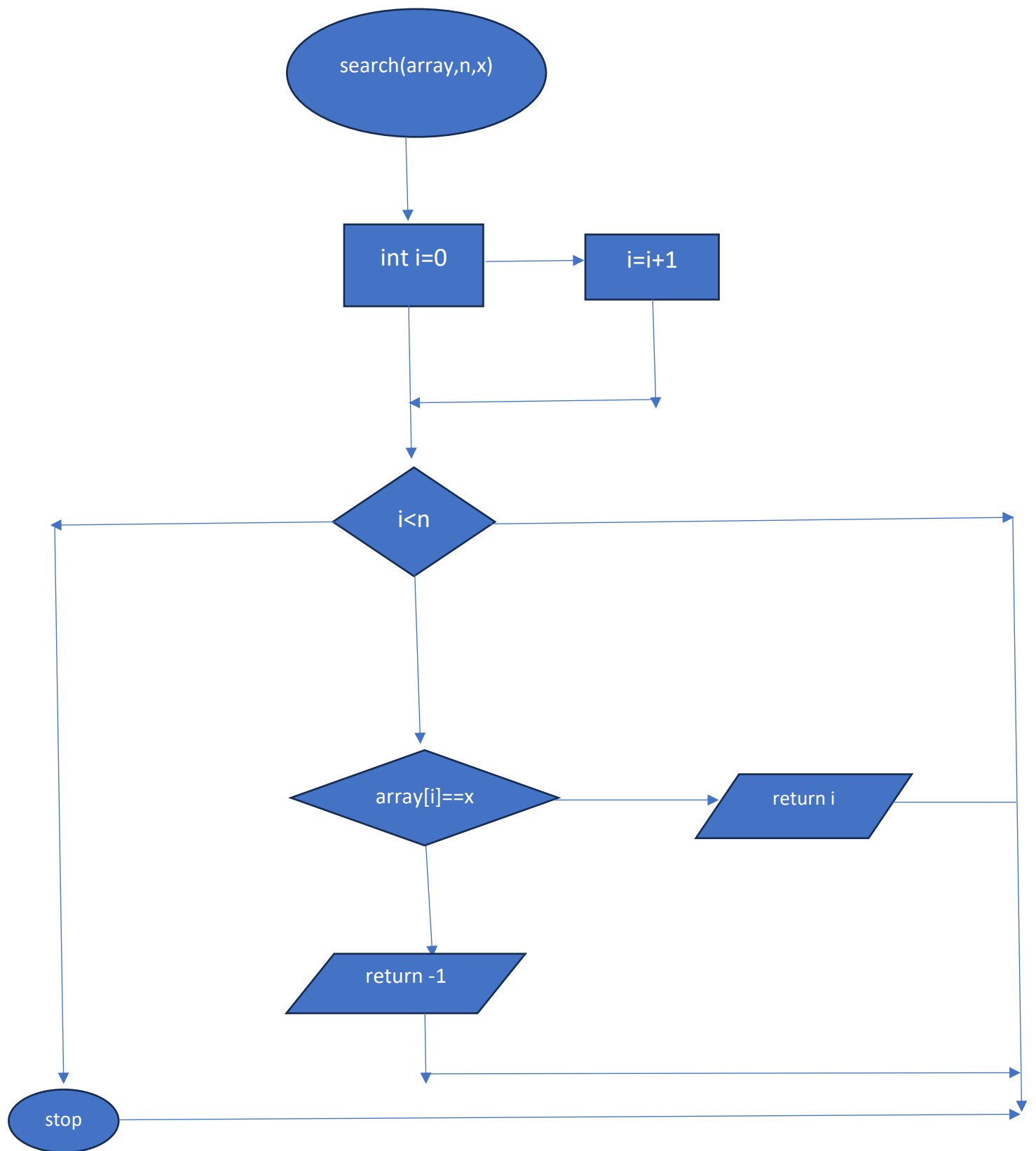
```
7 C:\mp7\0712\1\END.0
Enter the number of elements:5
Enter Element at array[0]:4
Enter Element at array[1]:2
Enter Element at array[2]:5
Enter Element at array[3]:0
Enter Element at array[4]:-1
Enter the element to be searched:5
5 is at index 2
```

ALGORITHM

- Start
- Declare array[] and variables i=0 and result ,x
- Input the elements of array and enter the number to be searched
- If(array[i]==x)
 - Return i
- Else
 - Return -1
- If result==-1
 - Print element not found
- Else
 - Print element at index i
- Stop

FLOWCHART





25: WRITE A PROGRAM TO SORT THE ELEMENTS OF THE ARRAY USING THE BUBBLE SORTING TECHNIQUE.

```
#include<stdio.h>

int bubblesort(int array[],int n)
{
    for(int i=0;i<=n;i++)
    {
        for(int j=0;j<n-i-1;j++)
        {
            if(array[j]<array[j+1])
            {
                int temp=array[j];
                array[j]=array[j+1];
                array[j+1]=temp;
            }
        }
    }
}

int main()
{
    int array[20],n,num;
    printf("Enter the number of elements:");
    scanf("%d",&n);
```

```

for(int i=0;i<n;i++)
{
    printf("Enter the element at array[%d]",i);
    scanf("%d",&array[i]);
}
num=bubblesort(array,n);
printf("The sorted array is:\n");
for(int i=0;i<n;i++)
{
    printf("%d\t",array[i]);
}
return 0;
}

```

OUTPUT

```

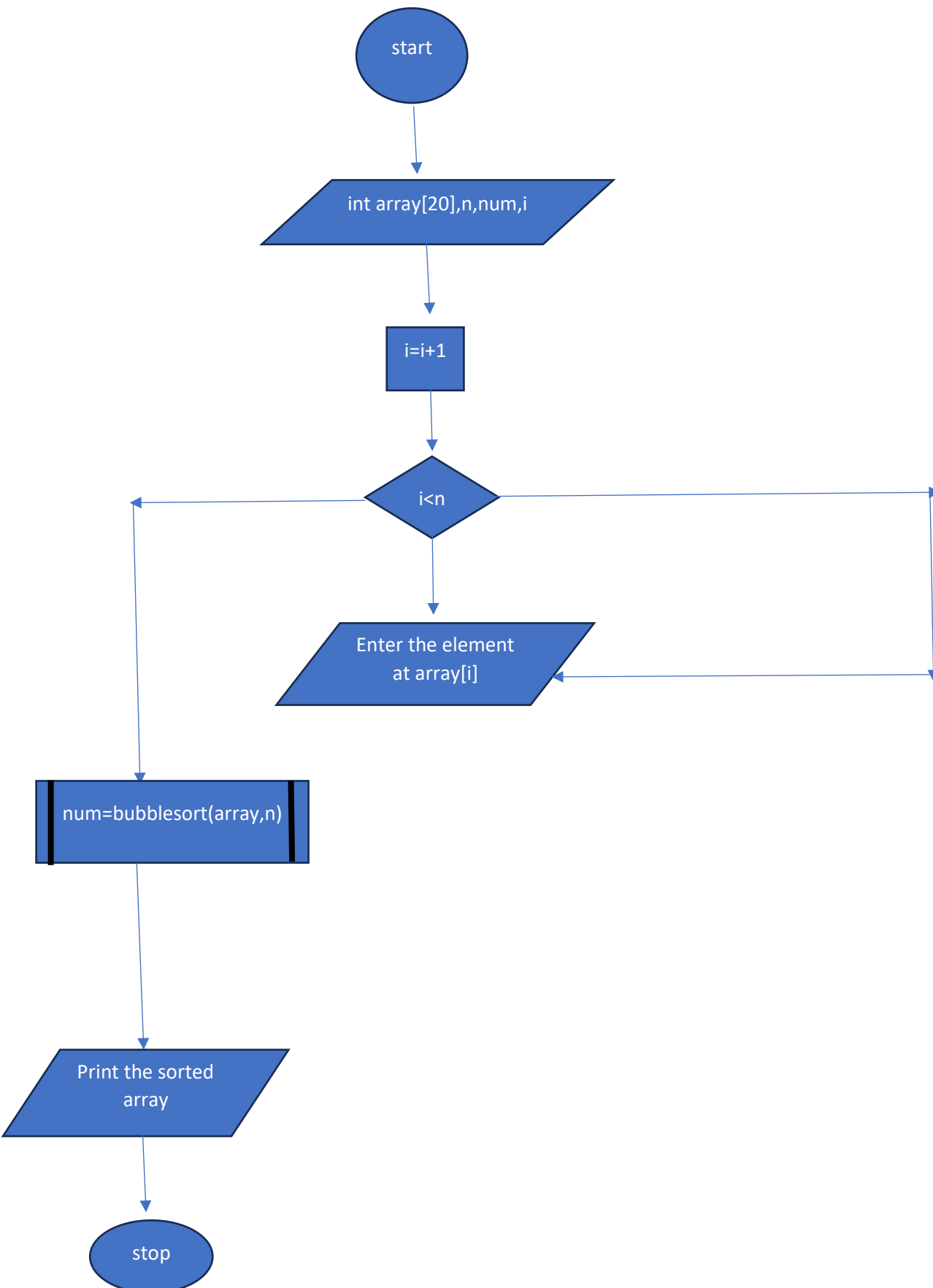
/tmp/U7rZxItcND.o
Enter the number of elements:5|
Enter the element at array[0]2
Enter the element at array[1]5
7Enter the element at array[2]
9
Enter the element at array[3]2
Enter the element at array[4]3
The sorted array is:
9    5    3    2    2

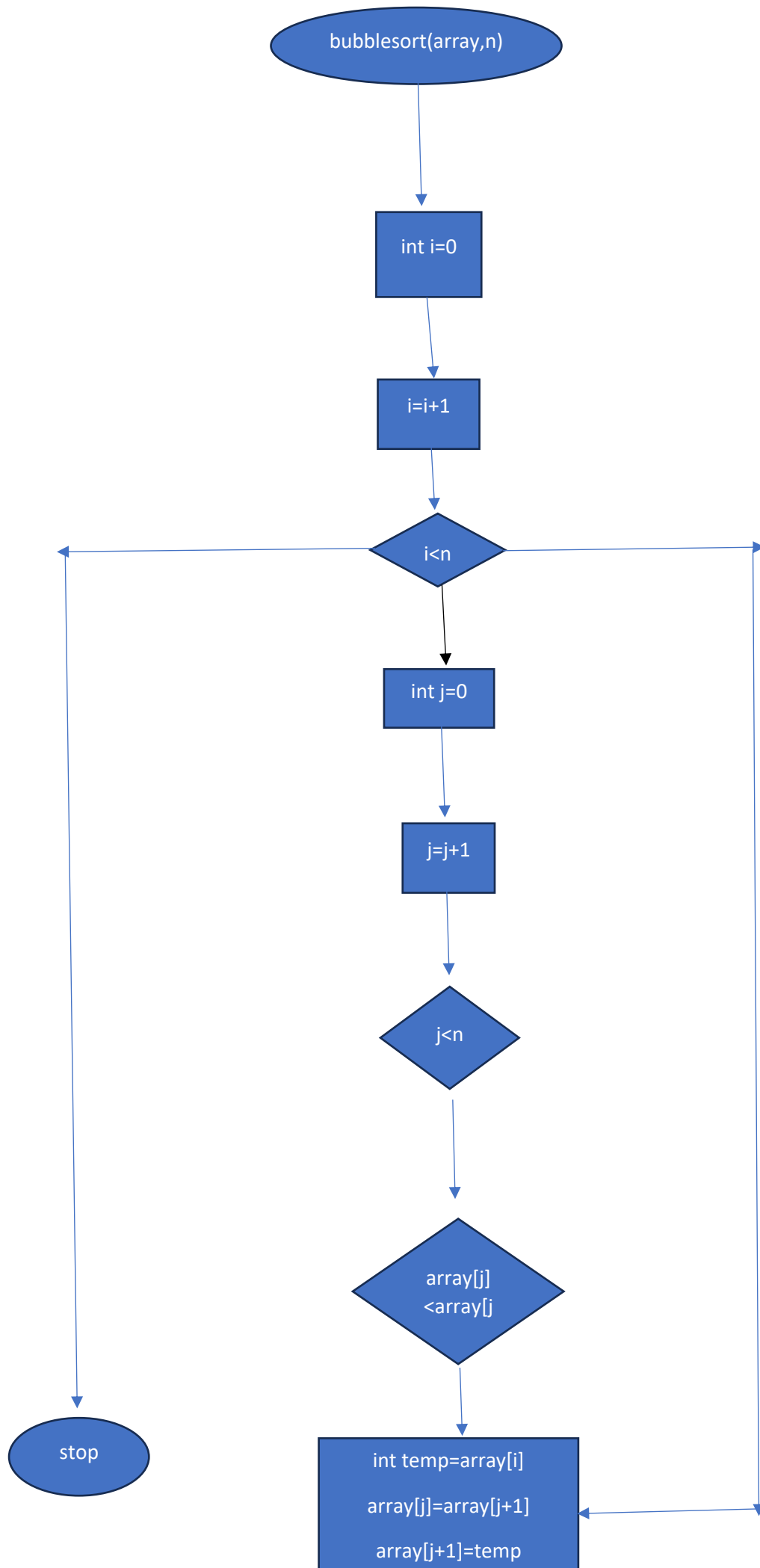
```

ALGORITHM

- Start
- bubblesort(array)
- enter the number of elements
- now starting with the first element index compare the current element with the next element of the array
- if the current element is greater than the next element of the array ,swap them
- if the current element is less than the next element move to the next element
- repeat the previous steps until the sorted array is obtained
- print the sorted array
- stop

FLOWCHART





26:WRITE A PROGRAM TO ADD AND MULTIPLY TWO MATRICES OF ORDER $n \times n$.

```
#include<stdio.h>

void main()
{
int a[5][5],b[5][5],c[5][5],d[5][5],i,j,k,rc1,sum=0;
printf("Enter the rows and columns of matrix 1 and matrix 2:");
scanf("%d",&rc1);
printf("Enter the elements of matrix 1:\n");
for(i=0;i<rc1;i++)
{
    for(j=0;j<rc1;j++)
    {
        printf("Enter the element at a[%d][%d]:",i,j);
        scanf("%d",&a[i][j]);
    }
}
printf("Enter the elements of matrix 2:\n");
for(i=0;i<rc1;i++)
{
    for(j=0;j<rc1;j++)
    {
        printf("Enter the element at b[%d][%d]:",i,j);
```

```

        scanf("%d",&b[i][j]);
    }
}
for(i=0;i<rc1;i++)
{
    for(j=0;j<rc1;j++)
    {
        sum=0;
        for(k=0;k<rc1;k++)
        {
            sum=sum+a[i][k]*b[k][j];
        }
        c[i][j]=sum;
    }
}
for(i=0;i<rc1;i++)
{
    for(j=0;j<rc1;j++)
    {
        d[i][j]=a[i][j]+b[i][j];
    }
}
printf("The addition of two matrix is:\n");
for(i=0;i<rc1;i++)

```

```
{
    printf("\n");
    for(j=0;j<rc1;j++)
    {
        printf("%d\t",d[i][j]);
    }
}
printf("\n");
printf("The multiplication of two matrix is:\n");
for(i=0;i<rc1;i++)
{
    printf("\n");
    for(j=0;j<rc1;j++)
    {
        printf("%d\t",c[i][j]);
    }
}
}
```

OUTPUT

```
/tmp/U7rZxItcND.o
Enter the rows and columns of matrix 1 and matrix 2:2
Enter the elements of matrix 1:
Enter the element at a[0][0]:1
Enter the element at a[0][1]:1
Enter the element at a[1][0]:1
Enter the element at a[1][1]:1
Enter the elements of matrix 2:
Enter the element at b[0][0]:1
Enter the element at b[0][1]:1
Enter the element at b[1][0]:1
Enter the element at b[1][1]:1
The addition of two matrix is:

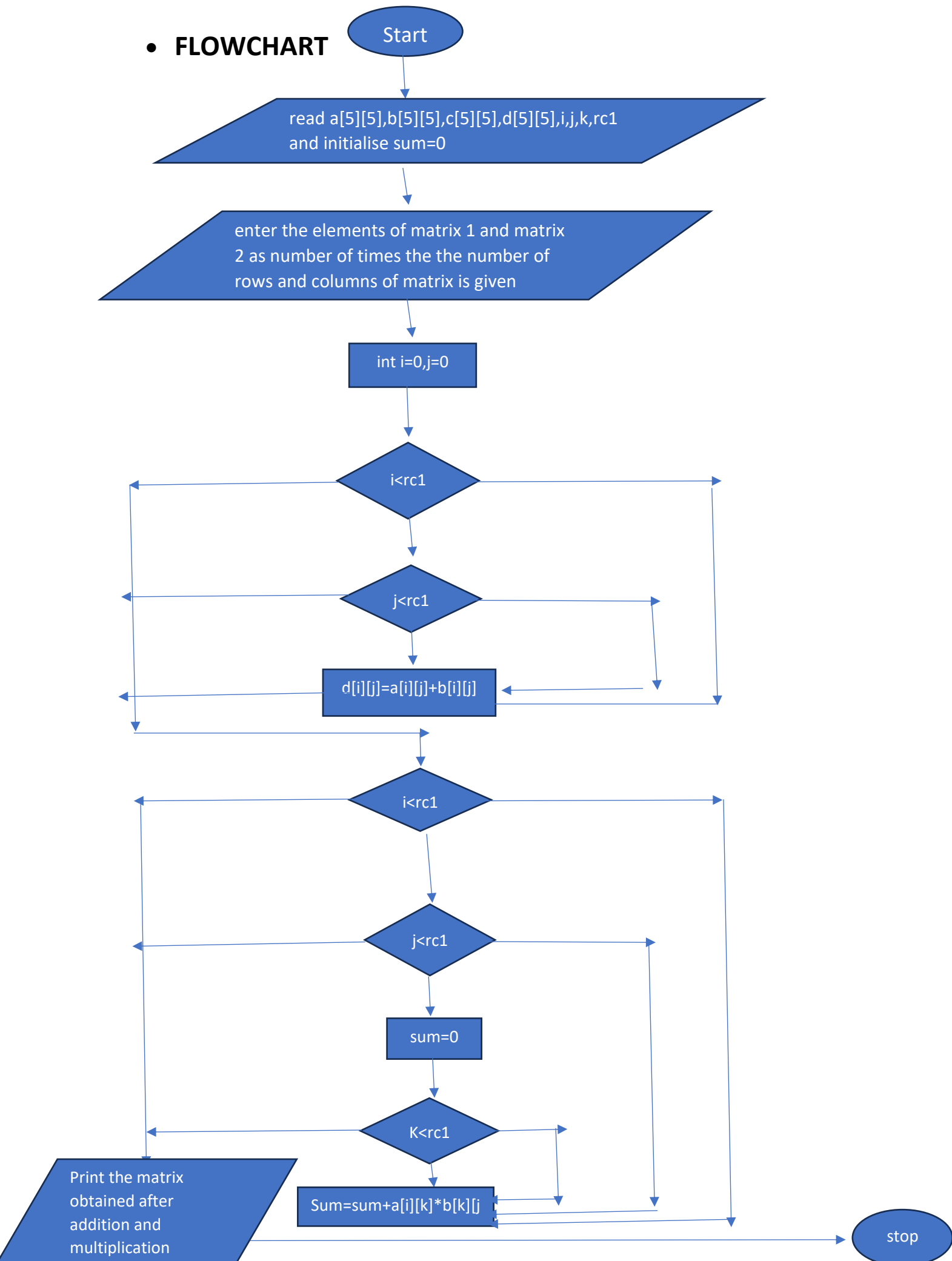
2  2
2  2
The multiplication of two matrix is:

2  2
2  2
```

ALGORITHM

- start
- read a[5][5],b[5][5],c[5][5],d[5][5],i,j,k,rc1 and initialise sum=0
- enter the elements of matrix 1 and matrix 2 as number of times the the number of rows and columns of matrix is given
- i=0 and j=0,sum=0 and k=0
sum=sum+a[i][k]*b[k][j]
until k<rc1 and
c[i][j]=sum
- repeat step 4 until i<rc1 And j<rc1
- i=0 and j=0
d[i][j]=a[i][j]+b[i][j] until j<rc1 and i<rc1
- print the matrix obtained after addition and multiplication.
- Stop

- **FLOWCHART**



27:WRITE A PROGRAM THAT FINDS THE SUM OF DIAGONAL ELEMENTS OF A mXn matrix.

```
#include<stdio.h>

void main()
{
int a[5][5],i,j,rc,trace=0;
printf("Enter the rows and columns of matrix:");
scanf("%d",&rc);
printf("Enter the elements of matrix:\n");
for(i=0;i<rc;i++)
{
for(j=0;j<rc;j++)
{
printf("Enter the element at a[%d][%d]:",i,j);
scanf("%d",&a[i][j]);
}
}
for(i=0;i<rc;i++)
{
for(j=0;j<rc;j++)
{
```

```
    if(i==j)
    {
        trace=trace+a[i][i];
    }
}
}

printf("The trace of matrix is: %d",trace);
}
```

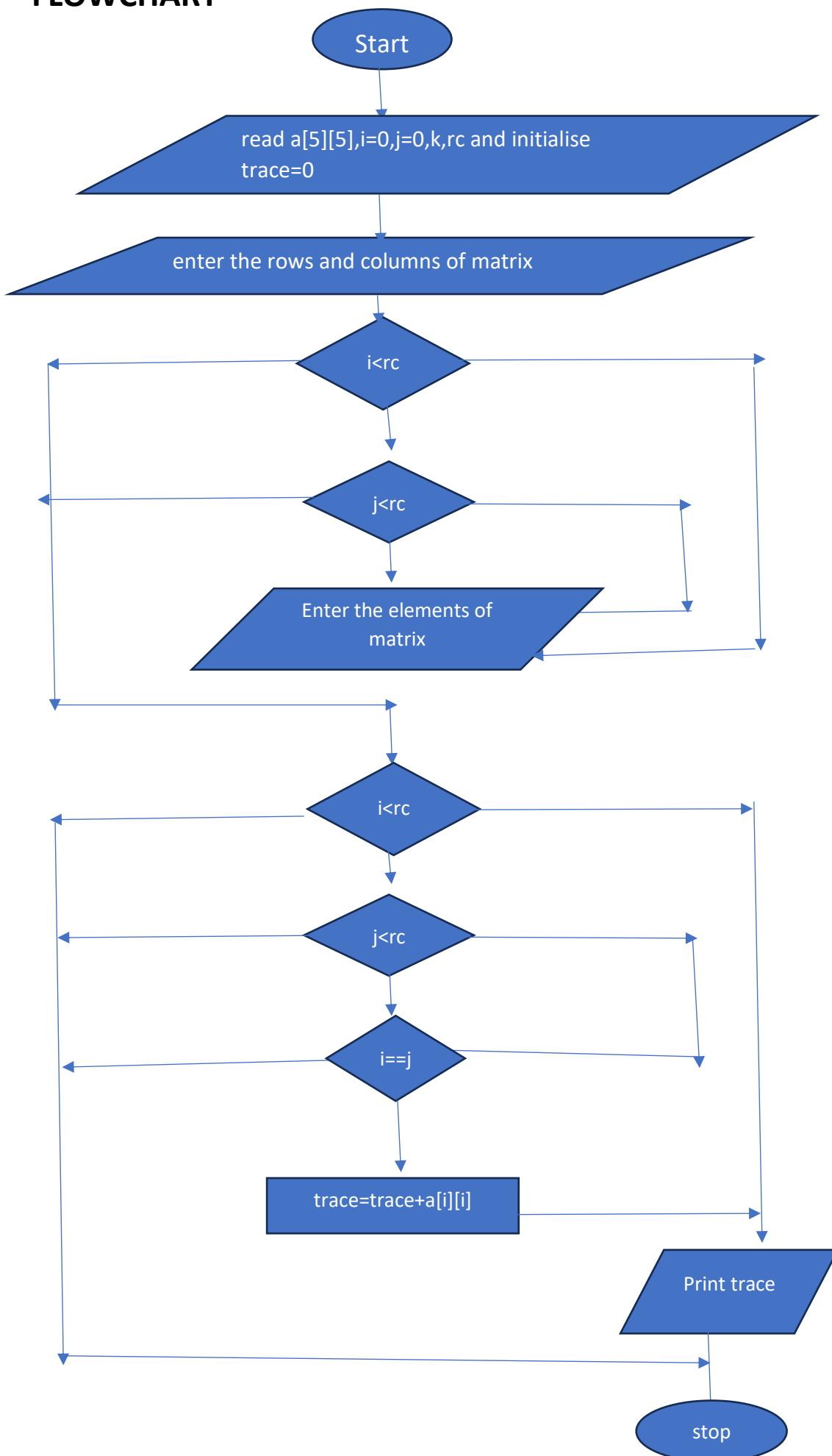
OUTPUT

```
/tmp/35yMT3ojAe.o
Enter the rows and columns of matrix:3
Enter the elements of matrix:
Enter the element at a[0][0]:1
Enter the element at a[0][1]:1
Enter the element at a[0][2]:1
Enter the element at a[1][0]:1
Enter the element at a[1][1]:1
Enter the element at a[1][2]:1
Enter the element at a[2][0]:1
Enter the element at a[2][1]:1
Enter the element at a[2][2]:1
The trace of matrix is: 3
```

ALGORITHM

- Start
- Read $a[5][5], i, j, rc, trace=0$
- Enter the rows and columns of matrix
- Enter the elements of matrix until $i < rc$ and $j < rc$
- If ($i==j$)
 $trace=trace+a[i][i]$
 and the loop continues until $i < rc$ and $j < rc$
- Print the trace of the matrix
- Stop

FLOWCHART

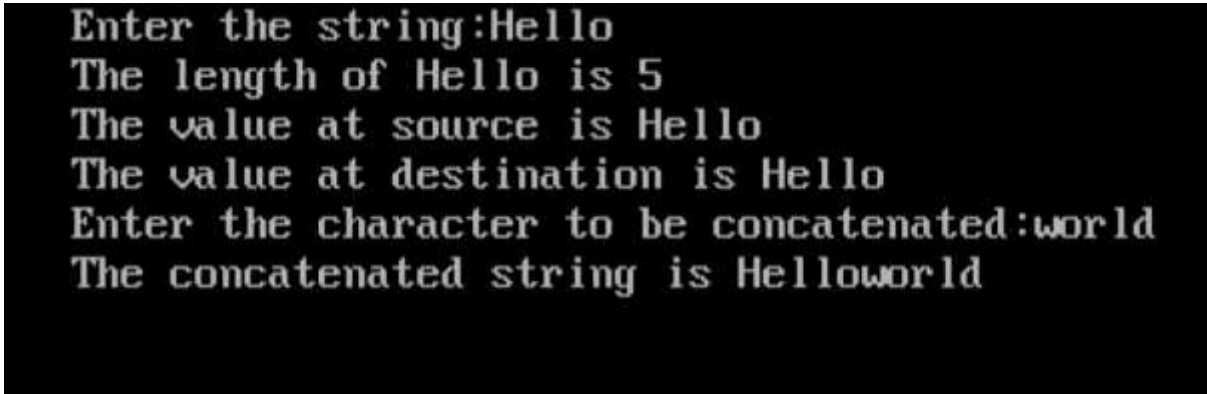


28:WRITE A PROGRAM TO IMPLEMENT strlen(),strcpy() and strcat().

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
#include<stdlib.h>
void main()
{
char string[50],dest[50],c1[50];
clrscr();
printf("Enter the string:");
gets(string);
printf("The length of %s is %d\n",string,strlen(string));
char dest[50];
strcpy(dest,string);
printf("The value at source is %s\n",string);
printf("The value at destination is %s\n",dest);
printf("Enter the character to be concatenated:");
scanf("%s",&c1);
strcat(string,c1);
printf("The concatenated string is %s\n",string);
```

```
getch();  
}
```

OUTPUT

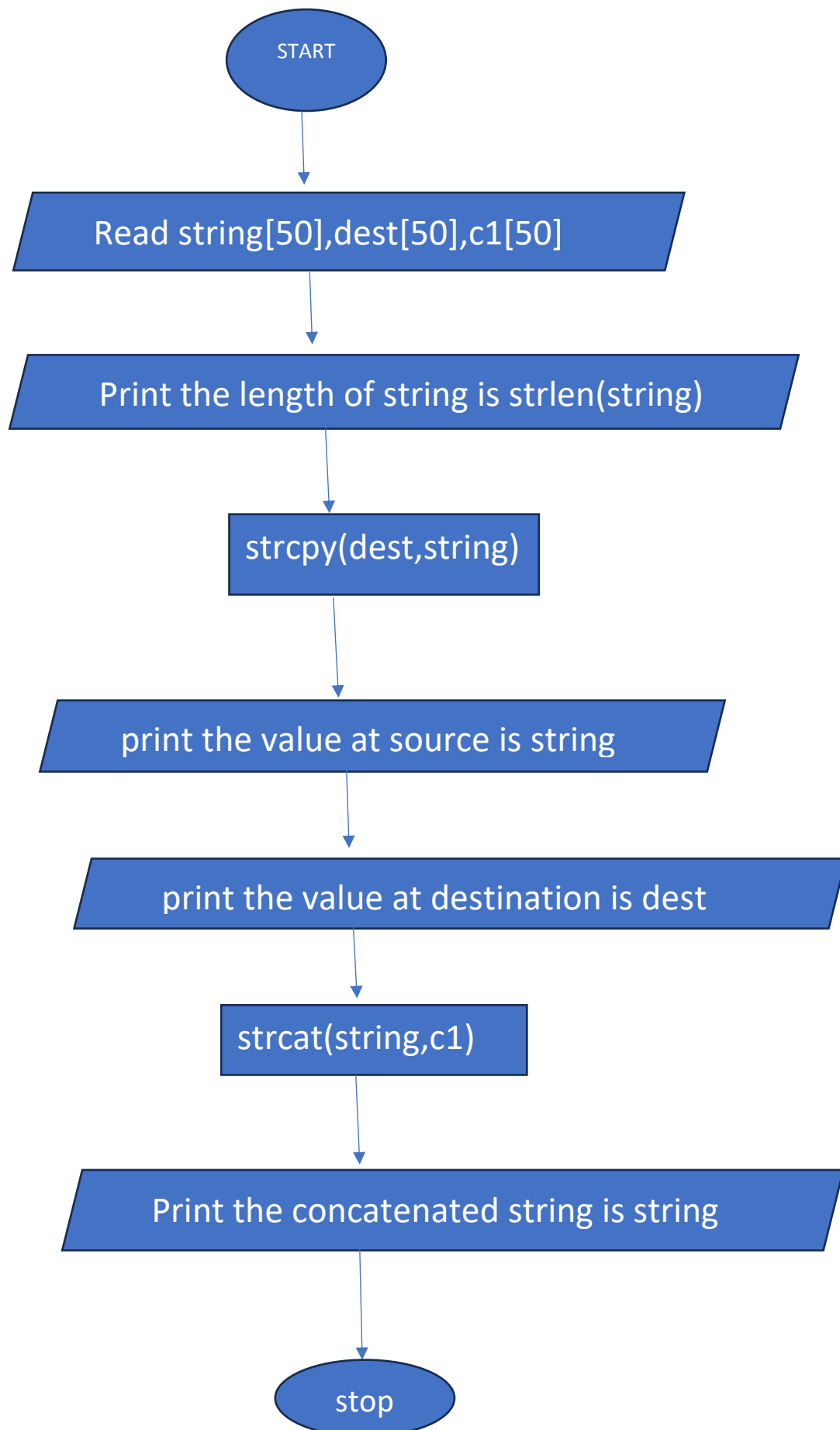
A screenshot of a terminal window with a black background and white text. The text shows the execution of a C program. It starts with 'Enter the string:Hello', followed by 'The length of Hello is 5', 'The value at source is Hello', 'The value at destination is Hello', 'Enter the character to be concatenated:world', and finally 'The concatenated string is Helloworld'.

```
Enter the string:Hello  
The length of Hello is 5  
The value at source is Hello  
The value at destination is Hello  
Enter the character to be concatenated:world  
The concatenated string is Helloworld
```

ALGORITHM

- Start
- Read string[50],dest[50],c1[50]
- Print the length of string is strlen(string)
- strcpy(dest,string)
- print the value at source is string
- print the value at destination is dest
- strcat(string,c1)
- print the concatenated string is string
- stop

FLOWCHART



29:WRITE A PROGRAM TO SWAP TWO ELEMENTS USING THE CONCEPT OF POINTERS.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int x,y,*ptr1,*ptr2,*ptr3;
clrscr();
printf("Enter the value of x:");
scanf("%d",&x);
printf("Enter the value of y:");
scanf("%d",&y);
ptr1=&x;
ptr2=&y;
printf("The value of x and y before swapping are %d and %d\n",*ptr1,*ptr2);
ptr3=ptr1;
ptr1=ptr2;
ptr2=ptr3;
printf("The value of x and y after swapping are %d and %d",*ptr1,*ptr2);
```

```
getch();  
}
```

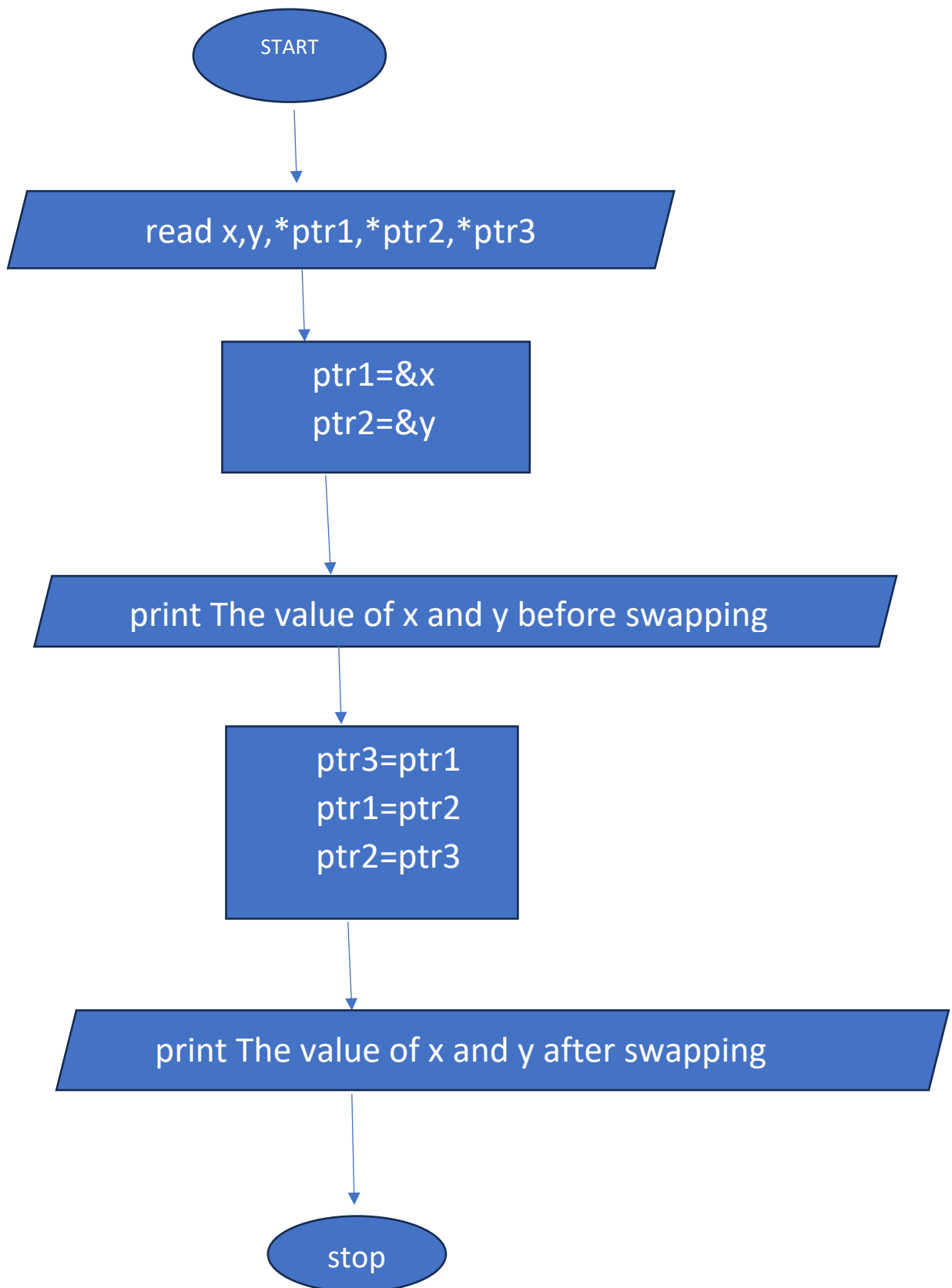
OUTPUT

```
Enter the value of x:19  
Enter the value of y:05  
The value of x and y before swapping are 19 and 5  
The value of x and y after swapping are 5 and 19_
```

ALGORITHM

- start
- read x,y,*ptr1,*ptr2,*ptr3;
- Enter the value of x and Enter the value of y
- ptr1=&x;
ptr2=&y;
- print The value of x and y before swapping
- ptr3=ptr1;
ptr1=ptr2;
ptr2=ptr3;
- printf The value of x and y after swapping
- stop

FLOWCHART



30: Define a structure data type TRAIN_INFO. The type contain Train No.: integer type Train name: string Departure Time: aggregate type TIME Arrival Time : aggregate type TIME Start station: string End station : string The structure type Time contains two integer members: hour and minute. Maintain a train timetable and implement the following operations: (i)List all the trains (sorted according to train number) that depart from a particular section. (ii)List all the trains that depart from a particular station at a particular time. (iii)List all the trains that depart from a particular station within the next one hour of a given time. (iv)List all the trains between a pair of start station and end station.

```
#include <stdio.h>
struct TRAIN{
int train_no;
char train_name[10];
    int arrhour;
    int arrmin;
    int departhour;
    int departmin;
    char start_station[20];
    char end_station[20];
} tr[10];
int main()
{
    int i,n,j,arr[10],h,m,station[20],bstation[20];
    printf("Enter the number of records to enter:");
    scanf("%d",&n);
    arr[n];
```

```

printf("Enter information of train:\n");
for (i=0;i<n;++i)
{
    printf("\nFor train number %d,\n",i+1);
    printf("Enter train number: ");
    scanf("%d", &tr[i].train_no);
    arr[i]=tr[i].train_no;
    printf("Enter train name: ");
    scanf("%s", &tr[i].train_name);
    printf("Enter start station: ");
    scanf("%s", &tr[i].start_station);
    printf("Enter end station: ");
    scanf("%s", &tr[i].end_station);
    printf("Enter arrival hour: ");
    scanf("%d", &tr[i].arrhour);
    printf("Enter arrival minute: ");
    scanf("%s", &tr[i].arrmin);
    printf("Enter departure hour: ");
    scanf("%d", &tr[i].departhour);
    printf("Enter departure minute: ");
    scanf("%d", &tr[i].departmin);
}
printf("Enter the hour:");
scanf("%d",&h);
printf("enter the minute:");
scanf("%d",&m);
for(i=0;i<n;i++)
{
    if(h==tr[i].departhour && m==tr[i].departmin)
    {

```

```

        printf("Train having departure time %d:%d are
%s\n",h,m,tr[i].train_name);
    }
    else
    {
        printf("No trains are departed at this time\n");
        break;
    }
}
for(int i=0;i<=n;i++)
{
    for(int j=0;j<n-i-1;j++)
    {
        if(arr[j]<arr[j+1])
        {
            int temp=arr[j];
            arr[j]=arr[j+1];
            arr[j+1]=temp;
        }
    }
}
printf("The sorted train number is:\n");
for(int i=0;i<n;i++)
{
    printf("%d\t",tr[i].train_no);
}
printf("\nEnter the end station:");
scanf("%s",&station);
for(i=0;i<n;i++)
{
    if(station==tr[i].end_station)

```

```

    {
        printf("%d train number is found ",tr[i].train_no);
    }
    else
    {
        printf("No train is found\n");
        break;
    }
}
printf("\nEnter the start station:");
scanf("%s",&bstation);
for(i=0;i<n;i++)
{
    if(station==tr[i].end_station &&
bstation==tr[i].start_station)
    {
        printf("train between %s and %s is
%d",tr[i].start_station,tr[i].end_station,tr[i].train_no);
    }
    else
    {
        printf("No train is found between these stations");
        break;
    }
}
return 0;
}

```

OUTPUT

```
/tmp/kaipp6f9r1.o
Enter the number of records to enter:1
Enter information of train:

For train number 1,
Enter train number: 12345
Enter train name: NDLS
Enter start station: VARANASI
Enter end station: DELHI
Enter arrival hour: 18
Enter arrival minute: 00
Enter departure hour: 18
Enter departure minute: 10
Enter the hour:18
enter the minute:09
No trains are departed at this time
The sorted train number is:
12345
Enter the end station:LUCKNOW
No train is found

Enter the start station:MUMBAI
No train is found between these stations
```

ALGORITHM

- start
- define structure train along with its members
- read i,n,j,arr[10],h,m,station[20],bstation[20]
- enter the information of train
- enter the various details regarding the train
until $i < n$
- if $n == tr[i].departhour$ and $m == tr[i].departmin$

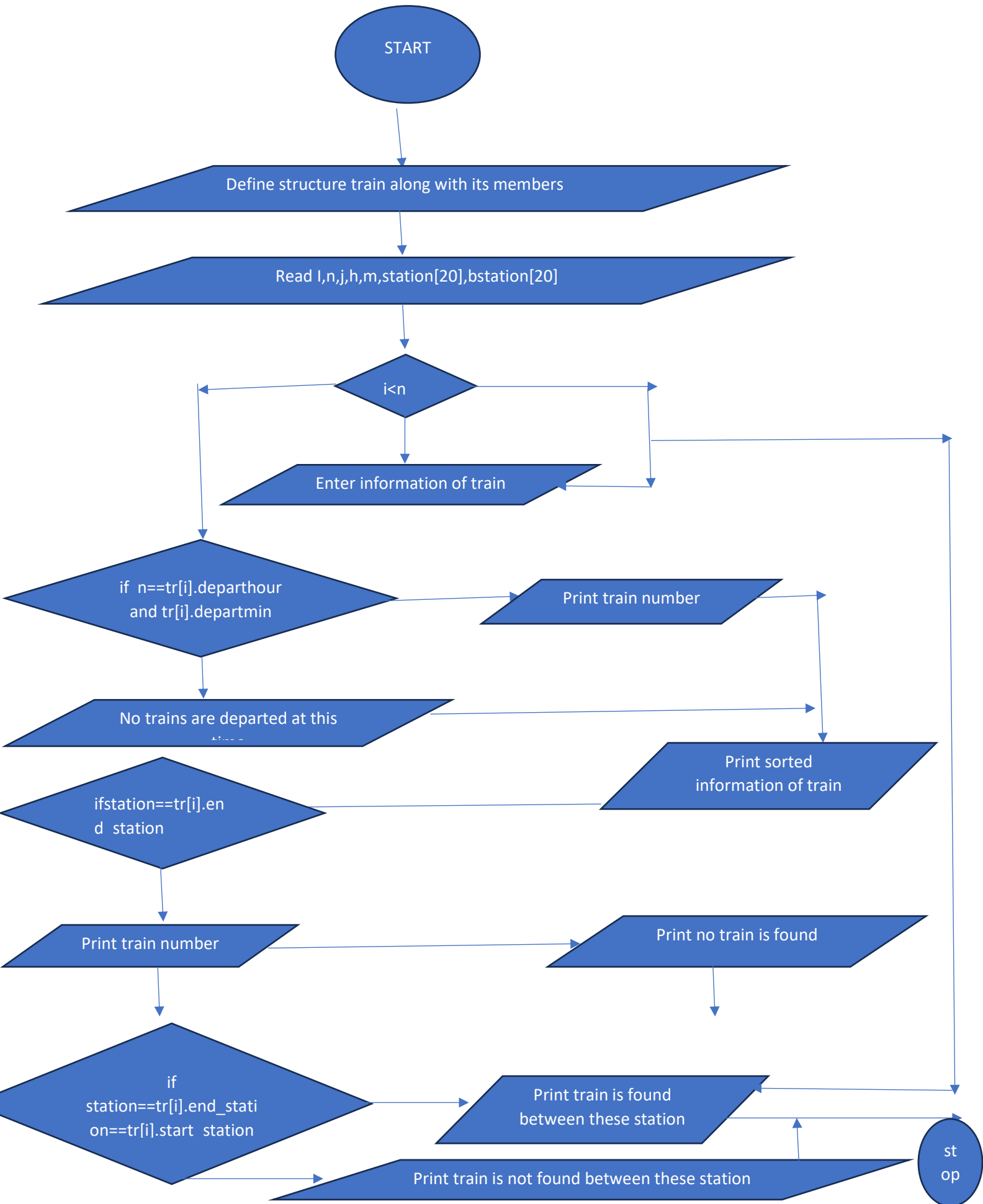
print the train number

else

print no trains are departed at this time

- sort the train information with respect to train no and print the sorted details ie the train no
- if(station==tr[i].end_station)
print train number is found
else
print train no is found
- if(station==tr[i].end_station and
bstation==tr[i].start_station)
print train is found between these stations
else
print no train is found between these stations
repeat these steps until $i < n$
- stop

FLOWCHART



31.WAP to compare the contents of two files and determine whether they are same or not.

```
#include <stdio.h>
#include <string.h>
int main (void)
{
    FILE *fptr1,*fptr2;
    char path [256];
    char string1 [20];
    char string2 [20];
    fptr1 = fopen("file1.txt", "r");
    if (fptr1 == NULL)
    {
        printf ("Error! Unable to open the file1.");
        return 1;
    }
    fptr2 =fopen("file2.txt", "r");
    if (fptr2 == NULL)
    {
```

```
    printf ("Error! Unable to open the file2.");
    fclose (fptr1);
    return 1;
}
while ((!feof (fptr1)) && (! feof (fptr2)))
{
    fscanf (fptr1, "%s", string1);
    fscanf (fptr2, "%s", string2);
    if (strcmp (string1, string2) != 0)
    {
        printf ("The content of files are not the same.");
    }
    else
    {
        printf("The content of files are not the same.");
    }
}
fclose (fptr1);
fclose (fptr2);
return 0;
```

}

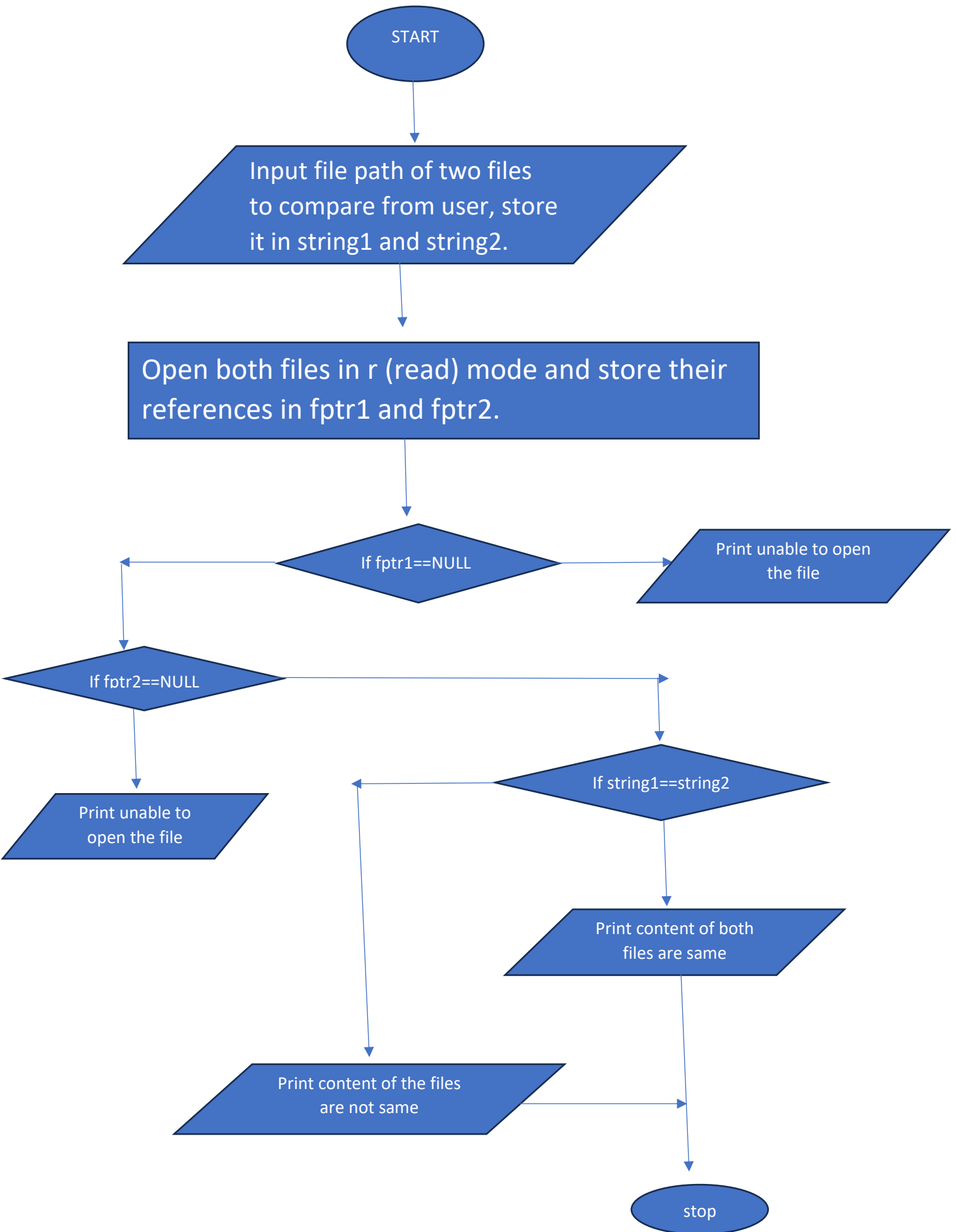
OUTPUT

```
/tmp/FvSzZjxqmN.o  
Error! Unable to open the file1.
```

ALGORITHM

- start
- Input file path of two files to compare from user, store it in string1 and string2.
- Open both files in r (read) mode and store their references in fptr1 and fptr2.
- If the file cannot be opened return sorry unable to return the file
- Else
Read a character from both files and compare.
If content of both files are same return that the contents are same
Else
Return the contents of file are not same
- Stop

FLOWCHART



32. WAP to check whether a given word exists in a file or not. If yes then find the number of times it occurs.

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
void main()
{
FILE*fp;
char path[255],targetword[100];
printf("Enter the path of the file:");
gets(path);
fp=fopen(path,"r");
if(fp==NULL)
{
printf("file not found!");
}
printf("Enter the word to be searched:");
scanf("%s",&targetword);
char current_word[100];
int ch,i=0;
int wordcounter=0;
```

```
while((ch=fgetc(fp))!=EOF)
{
    if(ch==" || ch=='\0')
    {
        current_word[i]='\0';
        i=0;
        if(!(strcmp(targetword,current_word)))
        {
            wordcounter++;
        }
    }
    else
    {
        currentword[i]=ch;
        i++;
    }
}
if(wordcounter==0)
{
    printf("word not found");
}
else
```

```
{  
    Printf("Matches found",wordcounter);  
}  
fclose(fp);  
return 0;  
}
```

OUTPUT

Enter the word to be searched:is

Word not found

ALGORITHM

- Start
- Read path[255],targetword[100],fp;
- Open the file
- if(fp==NULL)
 print file not found
 exit(1)
- read current_word[100],ch,i=0,wordcounter=0
- if(ch==" or ch=='\n')
 current_word[i]=='\n'
 if(strcmp(targetword,current_word)))
 wordcounter++
 else

current_word[i]=ch

i++

until end of file not found

- if(wordcounter==0)
print word not found
else
print matches found and wordcounter
- fclose(fp)
- stop

FLOWCHART

